

Project Data Page:





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Process Air 1st KOD 196-D

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Process Air Compressor 1st KO Drum-196-D -7%.pvdb

Date of Analysis : Sep 16,2026 2:23pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 4881.000 mm. ID
 Vessel Design Length, Tangent to Tangent 4575.00 mm.
 Distance of Bottom Tangent above Grade 3000.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 2700.00 mm.
 Internal Design Temperature 82 °C
 Internal Design Pressure 6.000 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 6 bars
 Shop Test Pressure 8.790 bars
 Field Test Pressure 7.800 bars
 Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -17.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-312 TP316L	...	...	S31603	
Nozzle	SA-182 F316L	...	<= 130	S31603	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F316L	...	<= 130	S31603	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt2	...	...	...	1.0000	N/A	No
Skirt1	...	...	...	1.0000	N/A	No
Bottom Clad Dis	6.182	0.00	7.000	1.0000	No	No
Clad shell 1	6.058	0.00	7.000	1.0000	N/A	No
Shell-1	6.000	0.00	7.000	1.0000	N/A	No
Shell-3	6.000	0.00	9.000	1.0000	N/A	No
Topdish	6.000	0.00	7.000	1.0000	No	No

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
---------	-----------	----------------	-------------------	--------------------	-------------------	-------------------	----------	----------

Type	mm.	mm.	mm.	mm.	mm.	mm.		
Skirt	-1000.0	1700.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	1000.0	12.0	12.0	...	...	0.70	0.50
Ellipse	50.0	50.0	18.0	15.0	13.9	...	0.85	0.85
Cylinder	1100.0	1050.0	16.0	16.0	13.6	...	0.85	0.85
Cylinder	2100.0	1000.0	16.0	16.0	13.5	...	0.85	0.85
Cylinder	4525.0	2425.0	20.0	20.0	13.5	...	0.85	0.85
Ellipse	4575.0	50.0	18.0	15.0	13.5	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.912	n/a	Passed
D1-2"	WRC-107/537	0.703	n/a	Passed
N1-36"	WRC-297	1.025	Nozzle	Failed
N2-36"	WRC-297	1.025	Nozzle	Failed

Wind/Earthquake Shear, Bending:

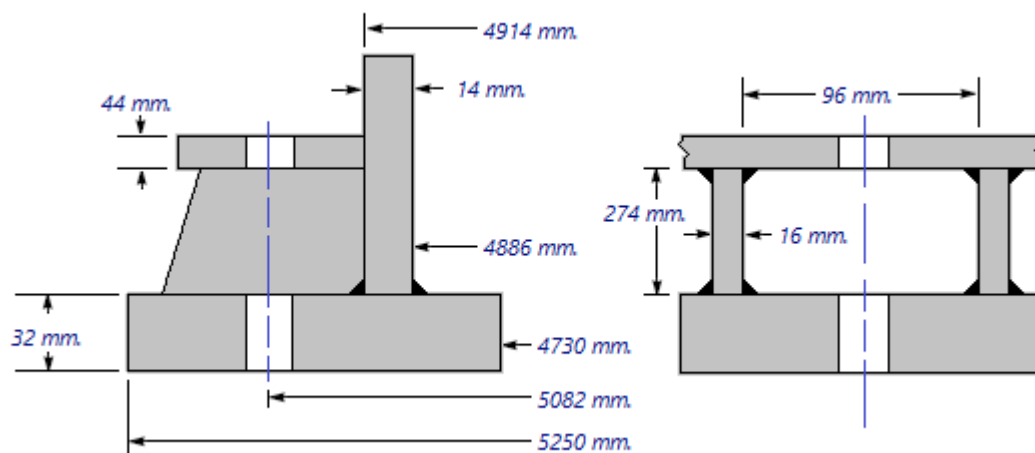
From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	2838.49	12334.9	13849.5	62020.9
20	30	2200	2446.99	12143	9356.79	41214.2
30	40	2725	2216.78	12001.4	7024.86	29141.7
40	50	3275	1949.61	8858	6920.7	28620.2
50	60	4300	1707.84	6964.98	5000.5	20313
60	80	6012.5	1477.59	6529.72	3407.75	13565.5
80	90	7250	918.324	3209.35	502.642	1756.63

Abs Max of the all of the Stress Ratio's : 0.4411

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	4730.0000	mm.
Outside Diameter of Basing	5250.0000	mm.
Nominal Diameter of Bolts	48.0000	mm.
Diameter of Bolt Circle	5082.0000	mm.
Number of Bolts	24	
Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	158.0000	mm.
Height of Gussets	274.0000	mm.
Distance between Gussets	96.0000	mm.
Thickness of Top Plate or Ring	44.0000	mm.
Radial Width of the Top Plate	168.0000	mm.
Number of Gussets	48	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	2838.	Kgf
Total Earthquake Shear on Support	12335.	Kgf
Wind Moment on Support	14550.	Kg-m.
Earthquake Moment on Support	62721.	Kg-m.

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	36181.7	kg.
Shop Test - Fabricated + water (full)	152590.1	kg.
Shipping - Fab. + rem. intls.+ shipping app.	36181.7	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	47145.9	kg.
Empty - Fab. + intls. + details + wghts.	47145.9	kg.
Operating - Empty + operating liquid (no ca)	74213.9	kg.
Field Test - Empty weight + test medium (full)	162763.2	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1-2"	6	...	...	OK	3.42	OK	No Calc[*]
K1-2"	6	...	...	OK	4.80	OK	No Calc[*]
K3-2"	6	...	...	OK	4.80	OK	No Calc[*]
K5-2"	6	...	...	OK	4.80	OK	No Calc[*]
K7-2"	6	...	...	OK	4.80	OK	No Calc[*]
K2-2"	6	...	...	OK	4.80	OK	No Calc[*]
K4-2"	6	...	...	OK	4.80	OK	No Calc[*]
K6-2"	6	...	...	OK	4.80	OK	No Calc[*]
K8-2"	6	...	...	OK	4.80	OK	No Calc[*]
K10-2"	6	...	...	OK	4.80	OK	No Calc[*]
N1-36"	6	...	...	OK	11.33	OK	Passed
N2-36"	6	...	...	OK	11.33	OK	Passed
M1-24"	6	...	...	...	...	OK	Passed
K9-2"	6	...	...	OK	5.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6 Nozzle : K9-2"
Minimum MAWP Shells/Flanges : 6 Element : Bottom Clad Dis
Minimum MAPnc Shells/Flanges : 7 Element : Topdish

Computed Vessel M.A.W.P. : 6 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	D1-2"	0.000	0.000	101.270
40	K1-2"	850.000	90.000	114.328
40	K3-2"	850.000	80.000	114.328
40	K5-2"	850.000	100.000	114.328
40	K7-2"	850.000	270.000	114.328
40	K2-2"	150.000	80.000	114.328
40	K4-2"	150.000	90.000	114.328
40	K6-2"	150.000	100.000	114.328
40	K8-2"	150.000	110.000	114.328
40	K10-2"	150.000	270.000	114.328
60	N1-36"	3100.000	90.000	1760.800
60	N2-36"	3500.000	270.000	1760.800
60	M1-24"	2650.000	180.000	1191.200
60	K9-2"	3650.000	110.000	122.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-2"	2.000 in	80S	WNF	2.375	5.537	...	...	165.10	600
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K7-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K8-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K10-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.74	300
K9-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.74	300
M1-24"	24.000 in	Actual	WNF	24.000	10.000	909.60	20.00	339.11	150
N1-36"	36.000 in	Actual	WNF	36.000	20.000	1614.40	20.00	470.00	150
N2-36"	36.000 in	Actual	WNF	36.000	20.000	1614.40	20.00	470.00	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-2"	SA-312 TP316L	15.000	10.000	...	...	...
K1-2"	SA-182 F316L	30.000	10.000	...	...	...
K3-2"	SA-182 F316L	30.000	10.000	...	...	...
K5-2"	SA-182 F316L	18.000	10.000	...	...	...
K7-2"	SA-182 F316L	30.000	10.000	...	...	...
K2-2"	SA-182 F316L	30.000	10.000	...	...	...
K4-2"	SA-182 F316L	30.000	10.000	...	...	...
K6-2"	SA-182 F316L	30.000	10.000	...	...	...
K8-2"	SA-182 F316L	30.000	10.000	...	...	...
K10-2"	SA-182 F316L	30.000	10.000	...	...	...
K9-2"	SA-105	20.000	10.000	...	...	...
M1-24"	SA-516 70	18.000	10.000	18.000	20.000	...
N1-36"	SA-516 70	16.000	10.000	16.000	20.000	14.000
N2-36"	SA-516 70	16.000	10.000	16.000	20.000	14.000

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-2"	...	0.0	150.00	0.00	Bottom Clad Dis
K1-2"	850.000	90.0	206.38	0.00	Clad shell 1
K3-2"	850.000	80.0	206.38	0.00	Clad shell 1
K5-2"	850.000	100.0	206.38	0.00	Clad shell 1
K7-2"	850.000	270.0	206.38	0.00	Clad shell 1
K2-2"	150.000	80.0	206.38	0.00	Clad shell 1
K4-2"	150.000	90.0	206.38	0.00	Clad shell 1
K6-2"	150.000	100.0	206.38	0.00	Clad shell 1

Process Air 1st KOD 196-D

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K8-2"	150.000	110.0	206.38	0.00	Clad shell 1
K10-2"	150.000	270.0	206.38	0.00	Clad shell 1
K9-2"	3650.000	110.0	206.38	0.00	Shell-3
M1-24"	2650.000	180.0	300.00	0.00	Shell-3
N1-36"	3100.000	90.0	300.00	150.00	Shell-3
N2-36"	3500.000	270.0	300.00	150.00	Shell-3

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm. THK X 4881.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3500.0mm	...
2	CLADDING: 1270.250mm X 3.000mm THK	...
1	CLADDING: 1050.000mm X 3.000mm THK	...
1	CLADDING: 1000.000mm X 3.000mm THK	...
1	CLADDING: 2425.000mm X 3.000mm THK	...
2	LIFTING LUGS:	SA-516 70
2	LIFTING LUG PADS: 700.0mm X 520.0mm X 20.0mm THK	
1	PLATFORM: 1000mm X 5000mm WIDE	
1	LADDER: 5600.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 5250.0mm. OD X 4730.0 ID X 32.0mm.	SA-516 70
24	BASERING BOLTS: 48.0mm. DIA	SA-193 B7
24	BASERING BOLT NUTS: 48.0mm. DIA	...
1	BASERING TOP RING: 5250.0mm. OD X 4914.0 ID X 44.0mm.	...
48	BASERING GUSSET PLATES: 158.0mm. X 158.0mm. X 274.0mm. X 16.0mm.	...
2	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 5044.0mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 2522.0mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 14.0mm. THK X 2739.6mm. X 1700.0mm.	SA-36
2	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 5046.0mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 2523.0mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 2740.8mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 16.0mm. THK X 5038.8mm. X 1050.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 2519.4mm. X 1050.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 2736.8mm. X 1050.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 16.0mm. THK X 3003.6mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 12330.2mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 1501.8mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 7509.1mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 20.0mm. THK X 5038.8mm. X 2425.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm. THK X 2519.4mm. X 2425.0mm.	SA-516 70

Process Air 1st KOD 196-D

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HEXAGON

1	CYLINDRICAL SEGMENT 20.0mm. THK X 2736.8mm. X 2425.0mm.	SA-516 70
1	NAMEPLATE	...

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**Minimum Design Metal Temperature Results Summary (°C):**

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Clad D[10]	B	-16	-29	-29	0.758	15.000	0.85	No
Bottom Clad Di[7]	B	-11	-31	-29	0.626	18.000	0.85	No
Clad shell 1 [8]	B	-14	-30	-29	0.710	16.000	0.85	No
Shell-1 [8]	B	-14	-30	-29	0.710	16.000	0.85	No
Shell-3 [8]	B	-7	-33	-29	0.560	20.000	0.85	No
Topdish [10]	B	-16	-29	-29	0.758	15.000	0.85	No
Topdish [7]	B	-11	-31	-29	0.626	18.000	0.85	No
N1-36" [1]	B	-7	-33	-29	0.560	20.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-36" [1]	B	-7	-33	-29	0.560	20.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
M1-24" [1]	B	-7	-33	-29	0.560	20.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K9-2" [1]	B	-13	-104	-29	0.007	16.764	1.00	No
Nozzle Flg [5]	!	-17	-17					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:		-7	-17					

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -17.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Process Air 1st KOD 196-D

Design Internal Pressure (for Hydrotest) 6 bars
 Design Internal Temperature 82.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -12.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-3
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(c)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.030
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt2
Distance "FROM" to "TO"	1700 mm.
Skirt Outside Diameter	4914 mm.
Diameter of Skirt at Base	4914 mm.
Skirt Thickness	14 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.25
Material Name	SA-36
Allowable Stress, Ambient	114 N./mm2
Allowable Stress, Operating	114 N./mm2
Allowable Stress, Hydrotest	225 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	230.08 N./mm2
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt1
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	4912 mm.
Diameter of Skirt at Base	4912 mm.
Skirt Thickness	12 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/7
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom Clad Dis
Distance "FROM" to "TO"	50 mm.
Inside Diameter	4881 mm.
Element Thickness	15 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	6 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm2
Platform Weight	5404.3 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3500 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-1220.2 mm.
Height/Length of Liquid	1270.2 mm.
Liquid Density	0.0009904 kg./cm3
Element From Node	30
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	-1220.2 mm.
Height/Length of Lining	1270.2 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3
Element From Node	30
Detail Type	Nozzle
Detail ID	D1-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	80S
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-312 TP316L
Element From Node	30
Detail Type	Weight



Detail ID	Skirt beam
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	600 Kg
Offset from Element Centerline	0 mm.

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Clad shell 1
Distance "FROM" to "TO"	1050 mm.
Inside Diameter	4881 mm.
Element Thickness	16 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	16 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	6 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	600 mm.
Liquid Density	0.0009904 kg./cm ³

Element From Node	40
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1050 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	40
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	80.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L



Element From Node 40
 Detail Type Nozzle
 Detail ID K5-2"
 Dist. from "FROM" Node / Offset dist 800 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 100.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F316L

Element From Node 40
 Detail Type Nozzle
 Detail ID K7-2"
 Dist. from "FROM" Node / Offset dist 800 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F316L

Element From Node 40
 Detail Type Nozzle
 Detail ID K2-2"
 Dist. from "FROM" Node / Offset dist 100 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 80.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F316L

Element From Node 40
 Detail Type Nozzle
 Detail ID K4-2"
 Dist. from "FROM" Node / Offset dist 100 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F316L

Element From Node 40
 Detail Type Nozzle
 Detail ID K6-2"
 Dist. from "FROM" Node / Offset dist 100 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 100.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F316L

Element From Node 40
 Detail Type Nozzle
 Detail ID K8-2"
 Dist. from "FROM" Node / Offset dist 100 mm.



Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	110.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K10-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Weight	
Detail ID	nozzles lining	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	20	Kgf
Offset from Element Centerline	20	mm.

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	1000	mm.
Inside Diameter	4881	mm.
Element Thickness	16	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	16	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	6	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.25	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	1000	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm3

Element#	6/7	
Element From Node	60	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	2425	mm.
Inside Diameter	4881	mm.



Element Thickness	20	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	20	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	6	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.25	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	0	mm.
Liquid Density	0	kg./cm3
Element From Node	60	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	2425	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm3
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N1-36"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N2-36"	
Dist. from "FROM" Node / Offset dist	1400	mm.
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	550	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	60	



Detail Type	Nozzle
Detail ID	K9-2"
Dist. from "FROM" Node / Offset dist	1550 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	110.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	60
Detail Type	Weight
Detail ID	Donut Ring
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	210 Kg
Offset from Element Centerline	200 mm.

Element From Node	60
Detail Type	Weight
Detail ID	Demister pad
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	410 Kg
Offset from Element Centerline	0 mm.

Element From Node	60
Detail Type	Weight
Detail ID	N1 Distributor
Dist. from "FROM" Node / Offset dist	1000 mm.
Miscellaneous Weight	2000 Kg
Offset from Element Centerline	1000 mm.

Element From Node	60
Detail Type	Weight
Detail ID	N2-Hopper
Dist. from "FROM" Node / Offset dist	1400 mm.
Miscellaneous Weight	1300 Kg
Offset from Element Centerline	1000 mm.

Element From Node	60
Detail Type	Weight
Detail ID	Davit
Dist. from "FROM" Node / Offset dist	550 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	1500 mm.

Element#	7/7
Element From Node	80
Element To Node	90
Element Type	Elliptical
Description	Topdish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	4881 mm.
Element Thickness	15 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	6 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0



Weld is pre-Heated	No
Element From Node	80
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	1300 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	72758 cm2
Platform Weight	4459.4 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	5000 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	2500 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	135.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	5600 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	5000 mm.
Element From Node	80
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1270.2 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3
Element From Node	80
Detail Type	Weight
Detail ID	Clip weight
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	660 Kg
Offset from Element Centerline	0 mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	Skirt2	...	1700	...	1700
	Skirt1	...	2700	...	1000
	Bottom Clad Dis	...	2750	...	50
	Clad shell 1	...	3800	...	1050
	Shell-1	...	4800	...	1000
	Shell-3	...	7225	...	2425
	Topdish	...	7275	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
	Skirt2	...	14	1	4914	...	1
	Skirt1	...	12	1	4912	...	2
	Bottom Clad Dis	6.1819	18	1	4881	117.3	3
	Clad shell 1	6.0583	16	1	4881	117.3	4
	Shell-1	6	16	1	4881	117.3	5
	Shell-3	6	20	1	4881	117.3	6
	Topdish	6	18	1	4881	117.3	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
	Skirt2	...	No Calc	No Calc	14	No Calc	1
	Skirt1	...	No Calc	No Calc	12	No Calc	2
	Bottom Clad Dis	6	6.54376	7.20474	15	13.8675	3
	Clad shell 1	6	7.12141	7.65965	16	13.6499	4
	Shell-1	6	7.17975	7.65965	16	13.5277	5
	Shell-3	6	9.08545	9.56519	20	13.5277	6
	Topdish	6	6.72562	7.20474	15	13.4888	7

Minimum 6 7

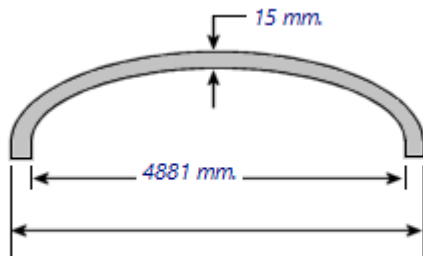
MAWP: 6 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Clad Dis

Elliptical Head Design Information:			
Design Pressure P	6.182 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	50.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	4881.000 mm.	Actual Stress	126.8 N./mm ²
Spec. Min. Thk t _e , or sa	15.0000 mm.	Spec. Nominal Thk.	18.000 mm.
Surface Area	269797.6 cm ²	SG of Contents	0.992
Element Volume	16160235.0 cm ³	Weight of Contents	15963.3 Kg
Empty Weight	4114.5 Kg	Operating Weight	20077.8 Kg
		Comp. All. Ope. Stress	85.5 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.013 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.506 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P)$$
 Appendix 1-4(c)

$$= (6.182 \cdot 4883.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 6.182)$$

$$= 12.8675 + 1.0000 = 13.8675 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.182 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t)$$
 per Appendix 1-4 (c)

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 14.0) / (0.999 \cdot 4883.0 + 0.2 \cdot 14.0)$$

$$= 7 - 0 = 7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t)$$
 per Appendix 1-4 (c)

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 15.0) / (1.0 \cdot 4881.0 + 0.2 \cdot 15.0)$$

$$= 7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.182 \cdot (0.999 \cdot 4883.0 + 0.2 \cdot 14.0)) / (2 \cdot 0.85 \cdot 14.0)$$

$$= 126.843 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (6.182 \cdot 2441.5) / (138.0 \cdot 0.85 - 0.6 \cdot 6.182) + 1.0$$

$$= 13.909 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.063 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 17.0) / (2441.5 + 0.6 \cdot 17.0)$$

$$= 8 - 0.1 = 8 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (4883.0 / (2 \cdot 1221.25)))^2 / 6$$

$$= 0.999454$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 1.609 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 15.0$, $t_r = 12.489$, $c = 1.0 \text{ mm.}$, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.758$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 18.0$, $t_r = 12.528$, $c = 1.0 \text{ mm.}$, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.626$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm.	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.626	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

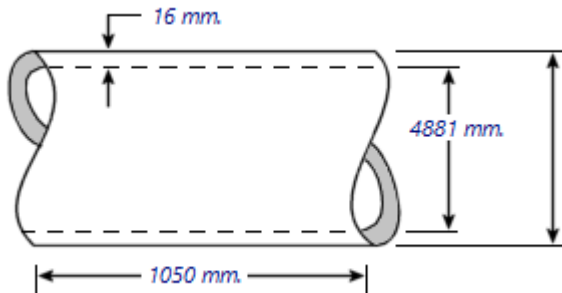
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.013 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.506 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Clad shell 1

Cylindrical Shell Design Information:			
Design Pressure P	6.058 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	1050.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	4881.000 mm.	Actual Stress	116.4 N./mm ²
Spec. Min. Thk t _e , or sa	16.0000 mm.	Spec. Nominal Thk.	16.000 mm.
Surface Area	162063.8 cm ²	SG of Contents	0.992
Element Volume	19650548.0 cm ³	Weight of Contents	11104.8 Kg
Empty Weight	2003.1 Kg	Operating Weight	13107.8 Kg
		Comp. All. Ope. Stress	91.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6.058 \cdot 2441.5) / (138.0 \cdot 0.85 - 0.6 \cdot 6.058) \\
 &= 12.6499 + 1.0000 = 13.6499 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.058 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 15.0) / (2441.5 + 0.6 \cdot 15.0) \\
 &= 7 - 0.1 = 7 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 16.0) / (2440.5 + 0.6 \cdot 16.0) \\
 &= 8 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6.058 \cdot (2441.5 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0) \\
 &= 116.446 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.327 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 16.0$, $t_r = 12.528$, $c = 1.0$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.71$, Temp. Reduction = $16\text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-14\text{ }^{\circ}\text{C}$

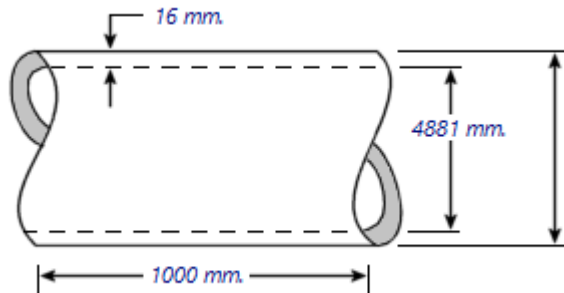
Min Metal Temp. at Required thickness (UCS 66.1) $-30\text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UG-20(f) $-29\text{ }^{\circ}\text{C}$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	$-12\text{ }^{\circ}\text{C}$
Governing Thk.	16.000 mm.	Basic MDMT	$-14\text{ }^{\circ}\text{C}$
Stress/Thk. Ratio	0.710	Temperature Difference	$16\text{ }^{\circ}\text{C}$
UCS-66(c) red.	No	Computed Min. Temp.	$-30\text{ }^{\circ}\text{C}$
		MDMT per UG-20(f)	$-29\text{ }^{\circ}\text{C}$
		Joint Efficiency E^*	85 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	6.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	4881.000 mm.	Actual Stress		115.3 N./mm^2
Spec. Min. Thk t,e, or sa		16.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		154346.5 cm^2	SG of Contents		0.000
Element Volume		18714806.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		1907.7 Kg	Operating Weight		1907.7 Kg
			Comp. All. Ope. Stress		91.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6.0 \cdot 2441.5) / (138.0 \cdot 0.85 - 0.6 \cdot 6.0) \\
 &= 12.5277 + 1.0000 = 13.5277 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 15.0) / (2441.5 + 0.6 \cdot 15.0) \\
 &= 7 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 16.0) / (2440.5 + 0.6 \cdot 16.0) \\
 &= 8 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6.0 \cdot (2441.5 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0) \\
 &= 115.324 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.327 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 12.528, c = 1.0 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.71$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

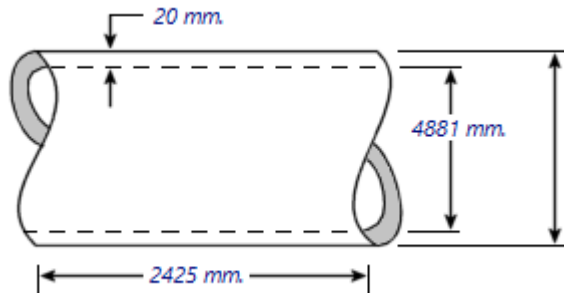
Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm.	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.710	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-30 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	6.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2425.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	4881.000 mm.	Actual Stress		91.1 N./mm^2
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		374899.6 cm^2	SG of Contents		0.000
Element Volume		45383404.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		5787.4 Kg	Operating Weight		5787.4 Kg
			Comp. All. Ope. Stress		101.0 N./mm^2
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6.0 \cdot 2441.5) / (138.0 \cdot 0.85 - 0.6 \cdot 6.0) \\
 &= 12.5277 + 1.0000 = 13.5277 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 19.0) / (2441.5 + 0.6 \cdot 19.0) \\
 &= 9 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 20.0) / (2440.5 + 0.6 \cdot 20.0) \\
 &= 10 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6.0 \cdot (2441.5 + 0.6 \cdot 19.0)) / (0.85 \cdot 19.0) \\
 &= 91.135 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.408 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20.0, t_r = 12.528, c = 1.0 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C

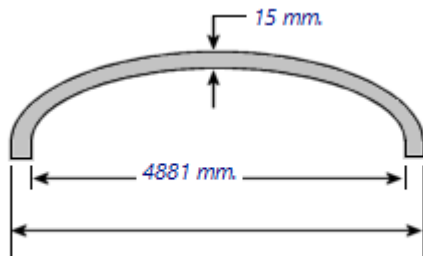
Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm.	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.560	Temperature Difference	26 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Topdish

Elliptical Head Design Information:					
Design Pressure	P	6.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	4881.000 mm.	Actual Stress		123.1 N./mm^2
Spec. Min. Thk t _e , or s _a		15.0000 mm.	Spec. Nominal Thk.		18.000 mm.
Surface Area		269797.6 cm^2	SG of Contents		0.000
Element Volume		16160235.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		4114.5 Kg	Operating Weight		4114.5 Kg
			Comp. All. Ope. Stress		85.5 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.013 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.506 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6.0 \cdot 4883.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 6.0)$$

$$= 12.4888 + 1.0000 = 13.4888 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 14.0) / (0.999 \cdot 4883.0 + 0.2 \cdot 14.0)$$

$$= 7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 15.0) / (1.0 \cdot 4881.0 + 0.2 \cdot 15.0)$$

$$= 7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.0 \cdot (0.999 \cdot 4883.0 + 0.2 \cdot 14.0)) / (2 \cdot 0.85 \cdot 14.0)$$

$$= 123.111 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (6.0 \cdot 2441.5) / (138.0 \cdot 0.85 - 0.6 \cdot 6.0) + 1.0$$

$$= 13.528 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 17.0) / (2441.5 + 0.6 \cdot 17.0)$$

$$= 8 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (4883.0 / (2 \cdot 1221.25))^2) / 6$$

$$= 0.999454$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.609 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 15.0, tr = 12.489, c = 1.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.758$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 18.0, tr = 12.528, c = 1.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.626$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm.	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.626	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.013 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.506 mm.

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 7.800 bars
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 7.800 bars
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 8.790 bars
 UG100 = 1.10 * M.A.W.P. * Sa/S 6.600 bars
 PED = max(1.43*DP, 1.25*DP*ratio) 8.580 bars
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 8.580 bars
 App 27-4 = M.A.W.P. 6.000 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Clad Dis	177.5	234.0	0.759	9.27
Clad shell 1	167.0	234.0	0.714	9.27
Shell-1	167.0	234.0	0.714	9.27
Shell-3	133.7	234.0	0.571	9.27
Topdish	177.5	234.0	0.759	9.27

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	101.29	1.135
K3-2"	Nozzle	115.00	101.29	1.135
K5-2"	Nozzle	115.00	101.29	1.135
K7-2"	Nozzle	115.00	101.29	1.135
K2-2"	Nozzle	115.00	101.29	1.135
K4-2"	Nozzle	115.00	101.29	1.135
K6-2"	Nozzle	115.00	101.29	1.135
K8-2"	Nozzle	115.00	101.29	1.135
K10-2"	Nozzle	115.00	101.29	1.135
N1-36"	Nozzle	138.00	138.00	1.000
N1-36"	Pad	138.00	138.00	1.000
N2-36"	Nozzle	138.00	138.00	1.000
N2-36"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Clad Dis	138.00	138.00	1.000
Clad shell 1	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Topdish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	5.40	153.00	0.035



K1-2"	1.96	153.00	0.013
K3-2"	1.96	153.00	0.013
K5-2"	1.96	153.00	0.013
K7-2"	1.96	153.00	0.013
K2-2"	1.96	153.00	0.013
K4-2"	1.96	153.00	0.013
K6-2"	1.96	153.00	0.013
K8-2"	1.96	153.00	0.013
K10-2"	1.96	153.00	0.013
N1-36"	20.82	234.00	0.089
N2-36"	20.82	234.00	0.089
M1-24"	27.88	234.00	0.119
K9-2"	1.96	225.00	0.009

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.800 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.800 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9.193 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.600 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	8.580 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8.580 bars
App 27-4	= M.A.W.P.	6.000 bars

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 6.0 • 1.0
 = 7.800 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm2 & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Clad Dis	175.2	234.0	0.749	8.54
Clad shell 1	161.7	234.0	0.691	8.41
Shell-1	159.7	234.0	0.683	8.31
Shell-3	124.7	234.0	0.533	8.21
Topdish	163.6	234.0	0.699	7.97

Stress ratios for Nozzle and Pad Materials (N./mm2):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	101.29	1.135
K3-2"	Nozzle	115.00	101.29	1.135
K5-2"	Nozzle	115.00	101.29	1.135
K7-2"	Nozzle	115.00	101.29	1.135
K2-2"	Nozzle	115.00	101.29	1.135
K4-2"	Nozzle	115.00	101.29	1.135
K6-2"	Nozzle	115.00	101.29	1.135
K8-2"	Nozzle	115.00	101.29	1.135
K10-2"	Nozzle	115.00	101.29	1.135
N1-36"	Nozzle	138.00	138.00	1.000
N1-36"	Pad	138.00	138.00	1.000
N2-36"	Nozzle	138.00	138.00	1.000
N2-36"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm2):

Description	Ambient	Operating	Ratio
Bottom Clad Dis	138.00	138.00	1.000
Clad shell 1	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Topdish	138.00	138.00	1.000

Minimum 1.000

**Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):**

Description	Ambient	Operating	Ratio
D1-2"	4.97	153.00	0.033
K1-2"	1.78	153.00	0.012
K3-2"	1.78	153.00	0.012
K5-2"	1.78	153.00	0.012
K7-2"	1.78	153.00	0.012
K2-2"	1.78	153.00	0.012
K4-2"	1.78	153.00	0.012
K6-2"	1.78	153.00	0.012
K8-2"	1.78	153.00	0.012
K10-2"	1.78	153.00	0.012
N1-36"	18.44	234.00	0.079
N2-36"	18.44	234.00	0.079
M1-24"	24.70	234.00	0.106
K9-2"	1.74	225.00	0.008

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2700	...	...	No Calc	No Calc
20	30	2700	...	...	No Calc	No Calc
30	40	No Calc	4911	14	0.00039594	39.5847
40	50	5388.5	4913	15	0.0002055	20.5449
50	60	5388.5	4913	15	0.0002055	20.5449
60	80	5388.5	4921	19	0.00029027	29.02
80	90	No Calc	4911	14	0.00039594	39.5847

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	15	No Calc	...	1.25377
40	50	16	No Calc	...	0.8363
50	60	16	No Calc	...	0.8363
60	80	20	No Calc	...	1.49387
80	90	15	No Calc	...	1.25377

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	5388.5	No Calc	No Calc	No Calc
50	60	5388.5	No Calc	No Calc	No Calc
60	80	5388.5	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	5022.82	...	4820.55	...	351.598
20	30	1431.73	...	1312.69	...	100.221
30	40	4114.5	16160235	3885.92	16179724	288.015
40	50	2003.07	19650548	1878.27	19666654	140.215
50	60	1907.67	18714806	1788.82	18730146	133.537
60	80	5787.39	45383404	5499.16	45420600	405.117
80	90	4114.5	16160235	3885.92	16179724	288.015
Total		24381	0.116E+09	23071	0.116E+09	1706

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	5782.59	...	...	0.29511E-04	PLAT:[1 OF 1]
30	Liqd	15963.3	...	-610.125	...	Liquid: 10
30	Lini	702.707	...	-585.125	...	clad
30	Nozl	6.74172	...	-1337.76	...	D1-2"
30	Wght	600	...	...	...	Skirt beam
40	Liqd	11104.8	...	300	...	Liquid: 20
40	Lini	386.628	...	525	...	clad
40	Nozl	11.5383	2543.69	800	...	K1-2"
40	Nozl	11.5383	2505.04	800	-441.707	K3-2"
40	Nozl	11.5383	2505.04	800	441.707	K5-2"
40	Nozl	11.5383	-2543.69	800	...	K7-2"
40	Nozl	11.5383	2505.04	100	-441.707	K2-2"
40	Nozl	11.5383	2543.69	100	...	K4-2"
40	Nozl	11.5383	2505.04	100	441.707	K6-2"
40	Nozl	11.5383	2390.28	100	869.992	K8-2"
40	Nozl	11.5383	-2543.69	100	...	K10-2"
40	Wght	20	...	...	-20	nozzles lining
50	Lini	368.217	...	500	...	clad
60	Liqd	...	...	...	...	Liquid: 60
60	Lini	892.927	...	1212.5	...	clad
60	Nozl	819.453	2590.5	1000	...	N1-36"
60	Nozl	819.453	-2590.5	1400	...	N2-36"
60	Nozl	439.379	...	550	2590.5	M1-24"
60	Nozl	11.2579	2390.28	1550	869.992	K9-2"
60	Wght	210	...	...	-200	Donut Ring
60	Wght	410	...	...	...	Demister pad
60	Wght	2000	1000	1000	...	N1 Distributor
60	Wght	1300	-1000	1400	...	N2-Hopper
60	Wght	50	...	550	1500	Davit
80	Plat	4771.54	...	1300	...	PLAT:[1 OF 1]
80	Lini	702.707	...	635.125	...	clad
80	Wght	660	...	...	...	Clip weight

Total Weight of Each Detail Type:

Platforms	10554.1
Liquid	27068.1
Cladding & Overlay	3053.2
Nozzles	2200.1
Weights	5250.0

Sum of the Detail Weights 48125.5 kg.

Weight Summation Results: (kg.)



	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	26088.4	26088.4	26088.4	26088.4	26088.4	26088.4
Cladding	3053.2	3053.2	3053.2	3053.2	3053.2	3053.2
Nozzles	2200.1	2200.1	2200.1	2200.1	2200.1	2200.1
Wld Weights	4840.0	4840.0	4840.0	4840.0	4840.0	4840.0
Platforms	...	...	...	10554.1	10554.1	10554.1
Empty Weights	...	...	...	410.0	410.0	...
Test Weights	...	410.0	...	...	...	...
Ope Weights	...	...	...	...	...	410.0
Ope. Liquid	...	...	...	...	...	27068.1
Test Liquid	...	115998.4	...	...	...	...
Totals	36181.7	152590.1	36181.7	47145.9	47145.9	74213.9

Field Installation Options:

Platforms installed before lifting.

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	36181.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	152590.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	36181.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	47145.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	47145.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	74213.9 kg.
Field Test Wt.	- Empty weight + test medium (full)	162763.2 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	22369.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	77456.3 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	22915.7 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	262442	260947
20	30	154315	153561
30	40	269798	265918
40	50	162064	161008
50	60	154346	153341
60	80	374900	371852
80	90	269798	265918

Total	1647662.4	1632544.8 cm ²
Total	1773.5	1757.3 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	5374.42	5374.42	5374.42	...	5374.42
20	30	1531.95	1531.95	1531.95	...	1531.95
30	40	5009.26	27457.9	27644.9	0.00017065	11494.6
40	50	2267.13	13758.5	22292.3	9.63843	2653.75
50	60	2041.21	2409.43	21112.8	...	2409.43

60	80	12252.1	13145	58500.7	1181.03	13145
80	90	5062.52	10536.8	26687.1	...	10536.8

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	47145.9	74213.9	163144
20	30	41771.4	68839.5	157770
30	40	40239.5	67307.5	156238
40	50	28744.9	39849.7	128593
50	60	26091.2	26091.2	106301
60	80	23681.7	23681.7	85187.8
80	90	10536.8	10536.8	26687.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1190.67	1190.67	1190.67
20	30	1190.67	1190.67	1190.67
30	40	1190.67	1190.67	1190.67
40	50	1190.67	1190.67	1190.67
50	60	1181.03	1181.03	1181.03
60	80	1181.03	1181.03	1181.03
80	90	...	...	...

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D1-2"	73.86	82.30	82	600	GR 2.3	...	...	...	...
K1-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K3-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K5-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K7-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K2-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K4-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K6-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K8-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K10-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
N1-36"	18.20	19.60	82	150	GR 1.1	9.589	18.204	18.204	17.717
N2-36"	18.20	19.60	82	150	GR 1.1	9.589	18.204	18.204	17.717
M1-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
K9-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	18.204	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1-36"	947.801	0.60	16348	0	6.000	18.204	Pass
N2-36"	947.801	0.60	16348	0	6.000	18.204	Pass

Evaluating Flange on Nozzle [N1-36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(16348) + 4 \cdot \text{abs}(0) \cdot 947.801 \leq \pi \cdot 947.801^3 \cdot ((18.204 - 6.000) + 0.600 \cdot 18.204)$$

$$= 0.26157E+06 \leq 0.63085E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.204, 0.600 \cdot 18.204 + 18.204 - 261570.8 / (\pi \cdot 947.801^3))$$

$$= \min(18.204, 19.538)$$

$$= 18.204 \text{ bars}$$

Evaluating Flange on Nozzle [N2-36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(16348) + 4 \cdot \text{abs}(0) \cdot 947.801 \leq \pi \cdot 947.801^3 \cdot ((18.204 - 6.000) + 0.600 \cdot 18.204)$$

$$= 0.26157E+06 \leq 0.63085E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.204, 0.600 \cdot 18.204 + 18.204 - 261570.8 / (\pi \cdot 947.801^3))$$

$$= \min(18.204, 19.538)$$

$$= 18.204 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition





The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 24.4145 Hz.

The Natural Frequency for the Vessel (Ope...) is 22.236 Hz.

The Natural Frequency for the Vessel (Filled) is 12.8232 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.936) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.891)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	22.236 Hz
Natural Frequency of Vessel (Empty)	24.415 Hz
Natural Frequency of Vessel (Test)	12.823 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.513
Structure Height to Diameter ratio	1.798
Height to top of Structure	8510.250 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850.0 + 300.0 = 1150.0$ mm.
 $= 3.773$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.773 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 $= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.513 \cdot 112.399$
 $= 784.6 \text{ lbs. } [355.9] \text{ Kg}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m2
Skirt2	1150.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt1	2500.0	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Clad Dis	3025.0	0.000	0.000	0.000	0.849	1.000	78.120
Clad shell 1	3575.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-1	4600.0	0.000	0.000	0.000	0.850	1.000	78.120
Shell-3	6312.5	0.000	0.000	0.000	0.909	1.000	78.120
Topdish	8072.3	0.000	0.000	0.000	0.957	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/7275.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[16.13/23.87^2] \cdot 10^4 = 283.098$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:
 $= W / (L \cdot Dr^2)$
 $= 72812 / (7275.0 \cdot 4918.306^2)$
 $= 0.41375\text{E-}06$

Since Vp is $> 0.400\text{E-}06$ no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	30000.00	3000.00	78.12	234.36	1.00
PLAT:[1 OF 1]	72758.41	8825.00	78.12	568.39	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Skirt beam	0.00	3000.00	78.12	0.00
nozzles linin	0.00	3050.00	78.12	0.00
Donut Ring	0.00	5100.00	78.12	0.00
Demister pad	0.00	5100.00	78.12	0.00
N1 Distribuor	0.00	6100.00	78.12	0.00
N2-Hopper	0.00	6500.00	78.12	0.00
Davit	0.00	5650.00	78.12	0.00
Clip weight	0.00	7525.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 22.236 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1150	6142.5	104422	78.12	391.507
20	30	2500	6140	61400	78.12	230.205
30	40	3025	6138.75	3069.38	78.12	267.174
40	50	3575	6141.25	64483.1	78.12	241.764
50	60	4600	6141.25	61412.5	78.12	230.252
60	80	6312.5	6151.25	149168	78.12	559.269
80	90	8072.34	6138.75	62625.3	78.12	918.324

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.030
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(27.87160.75)$
 $= 0.243$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/22.236)$
 $= 0.045$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.243, 0.045)$ per 12.8.2
 $= 0.045$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 2.5 \cdot 0.157 \cdot 74214(1 + 2 \cdot 0.03) / (1.25/1.25) \\
 &= 12334.947 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 74214 = 23273.49 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 74214$
 $= 4363.779 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 12334.947 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 74214$
 $= 2327.3 \text{ Kgf}$

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 22.236 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	850	5374.42	191.921	171.877	2327.35
20	30	2200	1531.95	141.592	126.804	2158.81
30	40	2725	27457.9	3143.44	513.578	2110.76
40	50	3275	13758.5	1893.02	279.353	1249.69
50	60	4300	2409.43	435.265	330.235	818.219
60	80	6012.5	13145	3320.37	2771.6	742.66
80	90	7250	10536.8	3209.35	1380.93	330.433

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	2838.49	12334.9	13849.5	62020.9
20	30	2200	2446.99	12143	9356.79	41214.2
30	40	2725	2216.78	12001.4	7024.86	29141.7
40	50	3275	1949.61	8858	6920.7	28620.2
50	60	4300	1707.84	6964.98	5000.5	20313
60	80	6012.5	1477.59	6529.72	3407.75	13565.5
80	90	7250	918.324	3209.35	502.642	1756.63

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	2838.49	0.0003722	0.0014019	0.15488E-05
20	30	2446.99	0.0022799	0.0033471	0.23057E-05
30	40	2216.78	0.0034049	0.0034631	0.23322E-05
40	50	1949.61	0.004752	0.0061573	0.27757E-05
50	60	1707.84	0.0075871	0.0090913	0.30733E-05
60	80	1477.59	0.012935	0.016927	0.33235E-05
80	90	918.324	0.01701	0.017093	0.33236E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	2451.9	15324.4
20	30	2450.9	15318.1
30	40	2450.4	15315
40	50	2451.4	15321.3
50	60	2451.4	15321.3
60	80	2455.39	15346.2
80	90	2450.4	15315

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 36.375 Actual deflection : 0.017 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	2155.11	2001.59	263252368	244599632
20	30	1847.27	1693.69	225738576	207054800
30	40	2307.18	2153.82	281540544	262932832
40	50	2461.49	2308.13	300370976	281770880
50	60	2461.49	2308.13	300370976	281770880
60	80	3079.38	2926.02	375773568	357203680
80	90	2307.18	2153.82	281540544	262932832

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-79.3472	-79.3472
20	30	82.7985	140.397	-67.1673	-67.1673
30	40	140.76	238.68	-85.5031	-85.5031
40	50	140.76	238.68	-91.5731	-91.5731
50	60	140.76	238.68	-91.5731	-91.5731
60	80	140.76	238.68	-101.046	-101.046
80	90	140.76	238.68	-85.5031	-85.5031

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	52.2009	...	67.8612
40	50	48.7128	...	63.3267
50	60	48.7128	...	63.3267
60	80	38.4322	...	49.9619
80	90	52.2009	...	67.8612

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-2.30992	-3.63612	-7.99327	0.047737	0.047737
20	30	-2.41866	-3.98596	-9.13522	0.056393	0.056393
30	40	-1.83219	-1.83219	-1.83219	0.044408	0.044408
40	50	-1.22132	-1.22132	-1.22132	0.041439	0.041439
50	60	-1.10857	-1.10857	-1.10857	0.041104	0.041104
60	80	-0.79372	-0.79372	-0.79372	0.032424	0.032424
80	90	-0.47976	-0.47976	-0.47976	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.047737	0.55526	2.48658	0.27763	...
20	30	0.056393	0.44316	1.95201	0.22158	...
30	40	0.044408	0.26201	1.0869	0.131	...
40	50	0.041439	0.24087	0.99609	0.12043	...
50	60	0.041104	0.17404	0.70696	0.087018	...
60	80	0.032424	0.093556	0.37243	0.046778	...
80	90	...	0.018747	0.065517	0.0093735	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	1.26522
20	30	...	...	...	1.05272
30	40	...	...	...	0.62986
40	50	...	...	...	0.57902
50	60	...	...	...	0.41009
60	80	...	...	...	0.20495
80	90	...	...	...	0.028191

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	0.11403
20	30	...	0.125
30	40	...	0.096108
40	50	...	0.053097
50	60	...	0.034765
60	80	...	0.024891
80	90	...	0.015045

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	80	...	...	...	...
80	90	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	80	...
80	90	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.91	79.35		0.0367
20		82.80	-2.92	67.17		0.0434
30		140.76	-2.14	85.50		0.0250
40		140.76	-1.50	91.57		0.0164
50		140.76	-1.32	91.57		0.0145
60		140.76	-0.92	101.05		0.0091
80		140.76	-0.50	85.50		0.0058

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.51	79.35		0.0442
20		82.80	-3.40	67.17		0.0507
30		140.76	-2.41	85.50		0.0282
40		140.76	-1.79	91.57		0.0195
50		140.76	-1.52	91.57		0.0167
60		140.76	-1.01	101.05		0.0100
80		140.76	-0.49	85.50		0.0058

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------



10	95.76	-4.24	79.35	0.0534
20	82.80	-4.49	67.17	0.0668
30	140.76	-2.14	85.50	0.0250
40	140.76	-1.50	91.57	0.0164
50	140.76	-1.32	91.57	0.0145
60	140.76	-0.92	101.05	0.0091
80	140.76	-0.50	85.50	0.0058

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-6.06	79.35		0.0763
20		82.80	-5.87	67.17		0.0874
30		140.76	-2.87	85.50		0.0335
40		140.76	-2.21	91.57		0.0241
50		140.76	-1.82	91.57		0.0199
60		140.76	-1.17	101.05		0.0116
80		140.76	-0.53	85.50		0.0062

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-8.32	79.35		0.1048
20		140.40	-9.41	67.17		0.1401
30		238.68	-2.01	85.50		0.0235
40		238.68	-1.38	91.57		0.0151
50		238.68	-1.24	91.57		0.0135
60		238.68	-0.87	101.05		0.0086
80		238.68	-0.49	85.50		0.0057

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-8.04	79.35		0.1013
20		140.40	-9.19	67.17		0.1368
30		238.68	-1.88	85.50		0.0219
40		238.68	-1.26	91.57		0.0138
50		238.68	-1.15	91.57		0.0126
60		238.68	-0.83	101.05		0.0082
80		238.68	-0.48	85.50		0.0056

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.24	79.35		0.0534
20		82.80	-4.49	67.17		0.0668
30	50.68	140.76		85.50	0.3600	
40	47.77	140.76		91.57	0.3394	
50	47.82	140.76		91.57	0.3397	
60	37.76	140.76		101.05	0.2683	
80	51.74	140.76		85.50	0.3676	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-6.06	79.35		0.0763
20		82.80	-5.87	67.17		0.0874
30	51.60	140.76		85.50	0.3666	
40	48.58	140.76		91.57	0.3451	
50	48.39	140.76		91.57	0.3438	
60	38.07	140.76		101.05	0.2704	
80	51.80	140.76		85.50	0.3680	



Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.24	79.35		0.0534
20		82.80	-4.49	67.17		0.0668
30		140.76	-2.14	85.50		0.0250
40		140.76	-1.50	91.57		0.0164
50		140.76	-1.32	91.57		0.0145
60		140.76	-0.92	101.05		0.0091
80		140.76	-0.50	85.50		0.0058

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-6.06	79.35		0.0763
20		82.80	-5.87	67.17		0.0874
30		140.76	-2.87	85.50		0.0335
40		140.76	-2.21	91.57		0.0241
50		140.76	-1.82	91.57		0.0199
60		140.76	-1.17	101.05		0.0116
80		140.76	-0.53	85.50		0.0062

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-8.32	79.35		0.1048
20		140.40	-9.41	67.17		0.1401
30	66.20	238.68		85.50	0.2774	
40	62.27	238.68		91.57	0.2609	
50	62.35	238.68		91.57	0.2612	
60	49.25	238.68		101.05	0.2063	
80	67.39	238.68		85.50	0.2823	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-8.04	79.35		0.1013
20		140.40	-9.19	67.17		0.1368
30	66.07	238.68		85.50	0.2768	
40	62.15	238.68		91.57	0.2604	
50	62.26	238.68		91.57	0.2608	
60	49.20	238.68		101.05	0.2061	
80	67.38	238.68		85.50	0.2823	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.36	79.35		0.0297
20		82.80	-2.48	67.17		0.0368
30	50.41	140.76		85.50	0.3581	
40	47.53	140.76		91.57	0.3377	
50	47.65	140.76		91.57	0.3385	
60	37.67	140.76		101.05	0.2676	
80	51.72	140.76		85.50	0.3674	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.38	79.35		0.0426
20		82.80	-3.65	67.17		0.0544
30	50.49	140.76		85.50	0.3587	
40	47.57	140.76		91.57	0.3379	
50	47.67	140.76		91.57	0.3387	



60	37.68	140.76	101.05	0.2677
80	51.75	140.76	85.50	0.3677

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.68	79.35		0.0464
20		82.80	-4.04	67.17		0.0602
30	50.41	140.76		85.50	0.3581	
40	47.53	140.76		91.57	0.3377	
50	47.65	140.76		91.57	0.3385	
60	37.67	140.76		101.05	0.2676	
80	51.72	140.76		85.50	0.3674	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.36	79.35		0.0297
20		82.80	-2.48	67.17		0.0368
30	50.41	140.76		85.50	0.3581	
40	47.53	140.76		91.57	0.3377	
50	47.65	140.76		91.57	0.3385	
60	37.67	140.76		101.05	0.2676	
80	51.72	140.76		85.50	0.3674	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.68	79.35		0.0464
20		82.80	-4.04	67.17		0.0602
30		140.76	-1.88	85.50		0.0219
40		140.76	-1.26	91.57		0.0138
50		140.76	-1.15	91.57		0.0126
60		140.76	-0.83	101.05		0.0082
80		140.76	-0.48	85.50		0.0056

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.57	66.12		0.0540
20		69.00	-3.92	55.97		0.0700
30	50.51	117.30		71.25	0.4306	
40	47.59	117.30		76.31	0.4057	
50	47.68	117.30		76.31	0.4065	
60	37.70	117.30		84.21	0.3214	
80	51.74	117.30		71.25	0.4411	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.57	66.12		0.0540
20		69.00	-3.92	55.97		0.0700
30		117.30	-1.78	71.25		0.0250
40		117.30	-1.21	76.31		0.0159
50		117.30	-1.11	76.31		0.0146
60		117.30	-0.80	84.21		0.0095
80		117.30	-0.46	71.25		0.0065

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.68	79.35		0.0464
20		82.80	-4.04	67.17		0.0602



30	50.41	140.76	85.50	0.3581
40	47.53	140.76	91.57	0.3377
50	47.65	140.76	91.57	0.3385
60	37.67	140.76	101.05	0.2676
80	51.72	140.76	85.50	0.3674

Absolute Maximum of the all of the Stress Ratio's 0.4411

Governing Element: Topdish

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt2	...	-3.5	...	...	...	+ Tens
Skirt2	...	-3.5	...	...	-3.6	- Comp
Skirt1	...	-3.9	0.1	...	...	+ Tens
Skirt1	...	-3.9	-0.1	...	-3.9	- Comp
Bottom Clad Dis	52.2	-1.7	...	...	50.5	+ Tens
Bottom Clad Dis	52.2	-1.7	...	...	...	- Comp
Clad shell 1	48.7	-1.2	...	...	47.6	+ Tens
Clad shell 1	48.7	-1.2	...	...	...	- Comp
Shell-1	48.7	-1.1	...	...	47.7	+ Tens
Shell-1	48.7	-1.1	...	...	...	- Comp
Shell-3	38.4	-0.8	...	...	37.7	+ Tens
Shell-3	38.4	-0.8	...	...	...	- Comp
Topdish	52.2	-0.5	...	...	51.7	+ Tens
Topdish	52.2	-0.5	...	...	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt1

Total vessel (empty) weight	Fg	40691.305 Kgf
Head Weight below section 2-2	DeltaFg	11494.557 Kgf
Weight of vessel contents (liquid)	Ff	27068.062 Kgf
Vessel Gross Weight	Fg + Ff	67759.359 Kgf
Skirt Mean Diameter	Dz	4901.000 mm.
Skirt Analysis Thickness (Corroded)	ez	11.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	29141.691 Kg-m.
Applied Moment - Bottom of Skirt	Mb	41214.160 Kg-m.
Weight at Bottom of Skirt	W	68839.500 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	4897.000 mm.
Head Corroded Thickness	eb	14.000 mm.

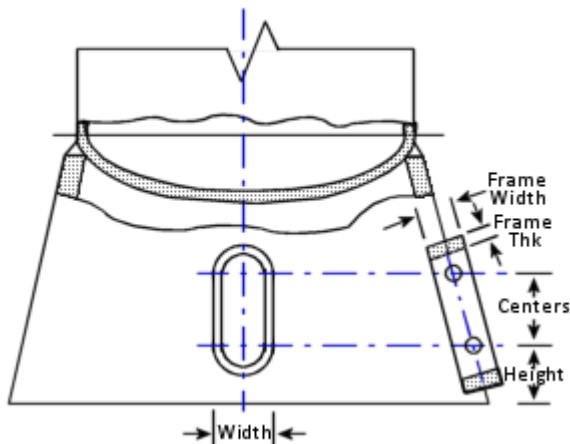
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 1221.25 \text{ mm.}$$

$$K = D_i / (2 * h_i) = 4883.0 / (2 * 1221.25) = 2.0$$

$$R = D_i (0.44 * K + 0.02) = 4883.0 * (0.44 * 2.0 + 0.02) = 4392.9404 \text{ mm.}$$

$$r = D_i ((0.5 / K) - 0.08) = 4883.0 * ((0.5 / 2.0) - 0.08) = 830.6102 \text{ mm.}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	50851788.000 cm ⁴
Moment of Inertia	Iyy =	50851788.000 cm ⁴
Section Modulus	W =	207516176.000 mm. ³
Section Metal Area	A =	1693.664 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})}$$

$$= \sqrt{(\min(50851788, 50851788) / 1693.664)}$$

$$= 1732.765 \text{ mm.}$$



Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 1732.765/1000.0$$

$$= 1.733$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 29141.689 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -29141.689/.83006E+09 - 68839.5/1693.66418$$

$$= -4.33 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +29141.689/.83006E+09 - 68839.5/1693.66418$$

$$= -3.642 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0.0 - 40691.305 - 27068.062 + 4 \cdot 29141.689/4901.0$$

$$= -43975.6 \text{ Kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0.0 - 40691.305 - 27068.062 - 4 \cdot 29141.689/4901.0$$

$$= -91543.2 \text{ Kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-43975.551 + 11494.557 + 27068.062) / (\pi \cdot 4897.0 \cdot 14.0) + 6.0 \cdot 4897.0 / (4 \cdot 14.0)$$

$$= 52.22 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-91543.172 + 11494.557 + 27068.062) / (\pi \cdot 4897.0 \cdot 14.0) + 6.0 \cdot 4897.0 / (4 \cdot 14.0)$$

$$= 50.06 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (27068.062 + 11494.557) / (\pi \cdot 4897.0 \cdot 14.0) + 6.0 \cdot 4897.0 / (4 \cdot 14.0)$$

$$= 54.23 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= -43975.551 / (\pi \cdot 4901.0 \cdot 11.0)$$

$$= -2.55 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -91543.172 / (\pi \cdot 4901.0 \cdot 11.0)$$

$$= -5.30 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(4897.0 + 14.0 - 4901.0 + 11.0) / (830.61 + 14.0) \\ = 0.9876$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.988) = 9.044 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ = 0.5 \cdot \sqrt{(14.0^2 + 11.0^2 + 2 \cdot 14.0 \cdot 11.0 \cdot 0.988)} \\ = 12.462 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 12.46 \cdot -43975.6 = -548.02 \text{ Kg-m.} \\ M_q = a \cdot F_{zq} = 12.46 \cdot -91543.2 = -1140.80 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2 \\ = 0.63 - 0.057(14.0/11.0)^2 \\ = 0.538$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ = 0.538 \cdot 6 \cdot -548.019 / (\pi \cdot 4897.0 \cdot 14.0^2) \\ = -5.75 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ = 0.538 \cdot 6 \cdot -1140.802 / (\pi \cdot 4897.0 \cdot 14.0^2) \\ = -11.97 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ = 0.538 \cdot 6 \cdot -548.019 / (\pi \cdot 4901.0 \cdot 11.0^2) \\ = -9.31 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ = 0.538 \cdot 6 \cdot -1140.802 / (\pi \cdot 4901.0 \cdot 11.0^2) \\ = -19.37 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.551/192.795 = 0.0029 \\ y = 125 \cdot eb/D_b = 125 \cdot 14.0/4897.0 = 0.3574$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994 \\ = 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.357 + 3.994 = 4.3285$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb) \\ = (6.0 + 0.0) \cdot 4897.0 ((9.04/40.0) \cdot 4.329 - 1) / (4 \cdot 14.0) \\ = -1.12 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ = 52.224 - -5.75 + -1.12 \\ = 56.85 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ = 52.224 + -5.75 - -1.12 \\ = 47.59 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 50.059 - -11.969 + -1.12$$

$$= 60.91 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 50.059 + -11.969 - -1.12$$

$$= 39.21 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 54.227 + -5.75 + -1.12$$

$$= 47.36 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 54.227 - -5.75 - -1.12$$

$$= 61.10 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 54.227 + -11.969 + -1.12$$

$$= 41.14 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 54.227 - -11.969 - -1.12$$

$$= 67.32 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -2.546 - -9.306$$

$$= 6.76 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -2.546 + -9.306$$

$$= -11.85 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -5.301 - -19.372$$

$$= 14.07 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -5.301 + -19.372$$

$$= -24.67 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138.0(3 - (1/1.5)) \cdot (52.224/137.997)^2$$

$$= 400.82 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138.0(3 - (1/1.5)) \cdot (50.059/137.997)^2$$

$$= 401.89 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138.0(3 - (1/1.5)) \cdot (54.227/137.997)^2$$

$$= 399.79 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (54.227/137.997)^2)$$

$$= 399.79 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-2.546/137.997)^2)$$

$$= 413.97 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-5.301/137.997)^2)$$

$$= 413.86 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	56.85	400.82	Pass	46
Stot1po	47.59	400.82	Pass	47
Stot1qi	60.91	401.89	Pass	48
Stot1qo	39.21	401.89	Pass	49
Stot2pi	47.36	399.79	Pass	50
Stot2po	61.10	399.79	Pass	51
Stot2qi	41.14	399.79	Pass	52
Stot2qo	67.32	399.79	Pass	53
Stot3pi	6.76	413.97	Pass	54
Stot3po	-11.85	413.97	Pass	55
Stot3qi	14.07	413.86	Pass	56
Stot3qo	-24.67	413.86	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	5333.494 mm.
Center of Gravity of the Liquid	2483.770 mm.
Center of Gravity of the Lining	4987.500 mm.
Center of Gravity of the Nozzles	5723.777 mm.
Center of Gravity of the Added Weights (Operating)	5589.904 mm.
Center of Gravity of the Added Weights (Empty)	5589.904 mm.
Center of Gravity of Bare Shell New and Cold	4020.275 mm.
Center of Gravity of Bare Shell Corroded	4015.997 mm.
Vessel CG in the Operating Condition	3843.184 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	4377.729 mm.
Vessel CG in the Test Condition	4850.739 mm.

Rigging Analysis Results:

• Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	7275.00 mm.
Total vessel weight (No Liquid)	Twt 47145.86 Kgf
Impact weight multiplication factor	Imp 1.50
Design lifting weight, DWT = Imp * Twt	70718.79 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	6915.00 mm.
Design Reaction force at the tailing lug	26523.69 Kgf
Design Reaction force at the lifting lug	44195.11 Kgf
CG distance during field lifting operation	4377.73 mm.
CG distance from tailing lug	4227.73 mm.
CG distance from the nearer lifting lug/trunnion	2537.27 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	1.58	2500.00	61.06 (UG-23)
Shear	1.04	6740.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	18405.6	0.8
1900.00	29469.6	1.3	977.6	0.1
3380.00	34000.5	1.1	5883.1	0.2
4600.00	30460.1	1.0	11238.5	0.4
7225.00	5394.3	0.2	1423.3	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0258 at elevation 2500.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 67 of 152

Center of Gravity Calculation: Step: 24 2:23pm Sep 16,2026



HEXAGON

--- Plot data successfully generated---

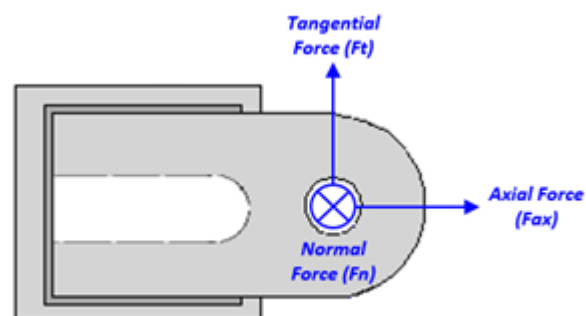
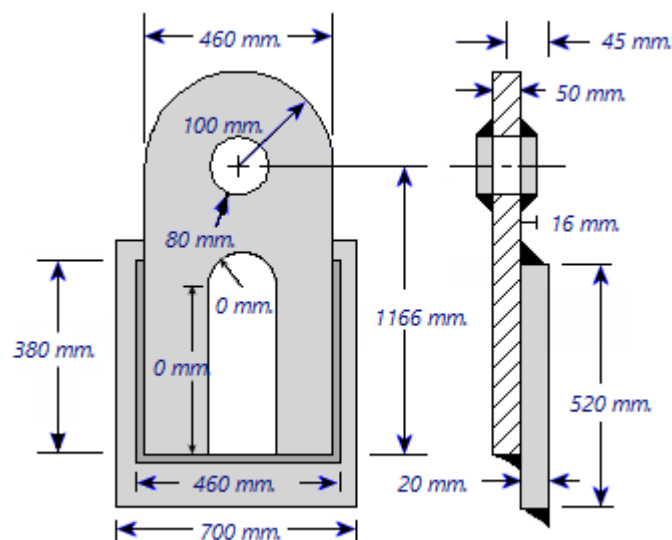
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Lifting Lug Calculations:

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Width of Lifting Lug	w	460.0000 mm.
Thickness of Lifting Lug	t	50.0000 mm.
Diameter of Hole in Lifting Lug	dh	80.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Bottom to Center of Hole	h	1166.0000 mm.
Offset from Vessel OD to Center of Hole	off	45.0000 mm.
Lug Fillet Weld Size	tw	20.0000 mm.
Length of weld along side of Lifting Lug	wl	380.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	460.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	700.00 mm.
Reinforcing Pad Longitudinal Length	wpl	520.00 mm.
Reinforcing Pad Thickness	thkp	20.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	16.00 mm.

Lifting Lug Orientation to Vessel: Flat
 Lift Orientation : From Horizontal to Vertical



PV Elite computes the forces and stresses from 0 to 90 degrees at 1 degree increments. The table below provides the results in 5 degree increments.

The 0 degree angle position is in the horizontal direction and the 90 degree angle position is in the vertical direction.

PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Lifting Lug and Weld Stresses at various Angles: N./mm²

Angle	----- Weld Stresses			----- Lug Stresses		Maximum
	Primary	Secondary	Allowable	Combined	Allowable	Ratio
0	16.0	92.3	104.8	96.6	172.9	0.881
5	17.1	93.8	104.8	99.0	172.9	0.895
10	18.2	94.5	104.8	100.6	172.9	0.902
15	19.3	94.3	104.8	101.3	172.9	0.900
20	20.3	93.4	104.8	101.2	172.9	0.891
25	21.3	91.6	104.8	100.2	172.9	0.874
30	22.3	88.9	104.8	98.3	172.9	0.849
35	23.3	85.5	104.8	95.6	172.9	0.816
40	24.2	81.4	104.8	92.1	172.9	0.777
45	25.0	76.5	104.8	87.8	172.9	0.730
50	25.9	70.8	104.8	82.8	172.9	0.676
55	26.7	64.5	104.8	76.9	172.9	0.616
60	27.5	57.5	104.8	70.3	172.9	0.549
65	28.3	49.8	104.8	63.1	172.9	0.475
70	29.2	41.5	104.8	55.1	172.9	0.396
75	30.1	32.5	104.8	46.4	172.9	0.310
80	31.2	22.7	104.8	36.9	172.9	0.298
85	32.6	12.0	104.8	26.6	172.9	0.311
90	34.5	2.5	104.8	15.1	172.9	0.329

Computed Results:

Total vessel weight (no liquid)	47145.86 Kg
Design reaction force at the tailing lug	26523.69 Kg
Design reaction force at the lifting lug	44195.11 Kg
Worst case lift angle for primary weld stresses	90 deg
Tangential force at this angle	0.0 Kg
Axial force at this angle	35359.4 Kg
Worst case lift angle for secondary weld stresses	12 deg
Tangential force at this angle	22632.9 Kg
Axial force at this angle	4810.8 Kg
Worst case lift angle for lug stresses	17 deg
Tangential force at this angle	22516.2 Kg
Axial force at this angle	6883.9 Kg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I _{ll}	28678.594 cm ⁴
Weld group centroid distance in the long. direction	Y _{ll}	264.305 mm.
Dist. of weld group centroid from lug bottom	Y _{llb}	115.695 mm.
Weld group inertia about the circumferential axis	I _{lc}	64894.500 cm ⁴
Weld group centroid distance in the circ. direction	Y _{lc}	230.000 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [S_{sl}]:

$$\begin{aligned}
 &= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot w_l + w_b) \cdot t_w) \\
 &= \sqrt{(35359^2 + 0^2 + 0^2)} / ((2 \cdot 380.0 + 460.0) \cdot 14.14) \\
 &= 20.10 \text{ N./mm}^2
 \end{aligned}$$



Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (F_n(h-Y_{lb}))Y_{ll}/I_{ll} + (F_{ax} \cdot \text{off} \cdot Y_{ll}/I_{ll}) + (F_t \cdot \text{off} \cdot Y_{lc}/I_{lc})$$

$$= (0(1166.0-115.695))264.305/28678.594 +$$

$$(35359 \cdot 45.0 \cdot 264.305/28678.594) +$$

$$(0 \cdot 45.0 \cdot 230.0/64894.5)$$

$$= 14.38 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= S_{sl} + S_{blf}$$

$$= 20.101 + 14.381$$

$$= 34.48 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Secondary Shear Stress in the Welds due to Shear Loads:

Unit Weld Polar Section Modulus [Uwsm]:

$$= (2 \cdot w_l + w)^3 / 12 - w_l^2 (w_l + w)^2 / (2 \cdot w_l + w)$$

$$= 67805392.00 \text{ mm}^3$$

Loads on Welds due to Torsional Moment:

Shear Stress in Normal Direction [Fth]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot (w_b/2) / U_{wsm}$$

$$= 80.43 \text{ Kg/mm.}$$

Shear Stress in Circumferential Direction [Ftv]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot \text{Cent} / U_{wsm}$$

$$= 91.49 \text{ Kg/mm.}$$

Primary Shear Stress [Fsv]:

$$= F_t / (2 \cdot w_l + w)$$

$$= 18.55 \text{ Kg/mm.}$$

Resultant Load on Weld Group [Fr]:

$$= \sqrt{(F_{th}^2 + (F_{tv} + F_{sv})^2)}$$

$$= \sqrt{(80.4^2 + (91.5 + 18.6)^2)}$$

$$= 136.30 \text{ Kg/mm.}$$

Resultant Secondary Weld Stress [Fws]:

$$= F_r / T_{weld}$$

$$= 136.304/14.14$$

$$= 94.53 \text{ N./mm}^2$$

Allowable Resultant Secondary Weld Stress [Psa]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262.0 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(P_l^2 + F_n^2)} / S_{ha}$$

$$= \sqrt{(35359^2 + 0^2)} / 60.0$$

$$= 57.79 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 50.0 + 2 \cdot 0.0$$

$$= 50.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(P_l^2 + F_n^2)} / (P_{bw} \cdot dh)$$

$$= \sqrt{(35359^2 + 0^2)} / (50.0 \cdot 80.0)$$

$$= 86.69 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= F_n \cdot (h - w_l) / (w \cdot t^2 / 6) + F_t \cdot (h - w_l) / (w^2 \cdot t / 6)$$

$$= 0 \cdot (1166.0 - 380.0) / (460.0 \cdot 50.0^2 / 6) +$$

$$22516 \cdot (1166.0 - 380.0) / (460.0^2 \cdot 50.0 / 6)$$

$$= 98.43 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_{ax} / (w \cdot t)$$

$$= 6884 / (460.0 \cdot 50.0)$$

$$= 2.94 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 98.4 + 2.9$$

$$= 101.36 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(P_l^2 + F_n^2))^{1/2} / (\pi \cdot S_{ta}))}$$

$$= \sqrt{(2(35359^2 + 0^2))^{1/2} / (\pi \cdot 105)}$$

$$= 45.8966 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

The full lug load is applied in both the circ. and long. directions simultaneously.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	4881.000 mm.
Vessel Thickness	Tv	20.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	500.00 mm.
Parameter C22	C22	400.00 mm.
Thickness of Reinforcing Pad	Tpad	20.000 mm.
Pad Parameter C11P	C11p	732.000 mm.
Pad Parameter C22P	C22p	536.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

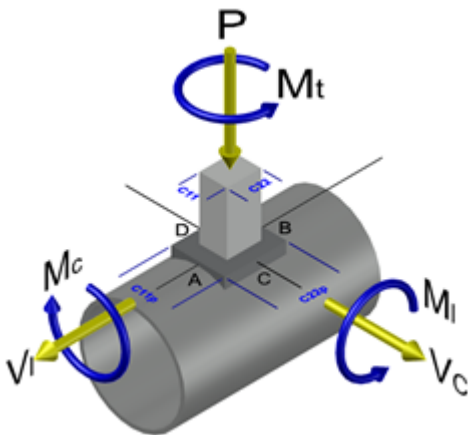
Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	35359.4	Kgf
Circumferential Shear (SUS)	Vc	35359.4	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.

Longitudinal Moment (SUS) Ml 0.0 Kg-m.
 Torsional Moment (SUS) Mt 34511.5 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	35359.4	Kgf
Longitudinal Shear	VL	35359.4	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	34511.5	Kg-m.

Dimensionless Parameters used : Gamma = 61.51

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.090	4C	10.465	(A,B)
$N(\Phi) / (P/R_m)$	0.090	3C	8.932	(C,D)
$M(\Phi) / (P)$	0.097	2C1	0.073	(A,B)
$M(\Phi) / (P)$	0.097	1C	0.109	(C,D)
$N(\Phi) / (MC/(R_m^{*2} * Beta))$	0.094	3A	2.039	(A,B,C,D)
$M(\Phi) / (MC/(R_m * Beta))$	0.105	1A	0.089	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{*2} * Beta))$	0.088	3B	6.800	(A,B,C,D)
$M(\Phi) / (ML/(R_m * Beta))$	0.091	1B	0.043	(A,B,C,D)
$N(x) / (P/R_m)$	0.096	3C	8.643	(A,B)
$N(x) / (P/R_m)$	0.096	4C	10.328	(C,D)
$M(x) / (P)$	0.092	1C1	0.116	(A,B)
$M(x) / (P)$	0.092	2C	0.077	(C,D)
$N(x) / (MC/(R_m^{*2} * Beta))$	0.094	4A	3.079	(A,B,C,D)
$M(x) / (MC/(R_m * Beta))$	0.106	2A	0.047	(A,B,C,D)
$N(x) / (ML/(R_m^{*2} * Beta))$	0.088	4B	1.894	(A,B,C,D)
$M(x) / (ML/(R_m * Beta))$	0.091	2B	0.066	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC		8.7	8.7	-8.7	-8.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-10.8	-10.8	10.8	10.8
Shear MT		169.2	169.2	169.2	169.2	169.2	169.2	169.2	169.2
Tot. Shear		177.9	177.9	160.6	160.6	158.4	158.4	180.1	180.1
Str. Int.		355.8	355.8	321.1	321.1	316.8	316.8	360.1	360.1

Dimensionless Parameters used : $\Gamma = 122.53$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.126	4C	16.393	(A,B)
N(PHI) / (P/Rm)	0.126	3C	10.026	(C,D)
M(PHI) / (P)	0.140	2C1	0.029	(A,B)
M(PHI) / (P)	0.140	1C !	0.068	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.135	3A	4.718	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.143	1A	0.067	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.121	3B	12.304	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.126	1B	0.022	(A,B,C,D)
N(x) / (P/Rm)	0.138	3C	8.973	(A,B)
N(x) / (P/Rm)	0.138	4C	15.691	(C,D)
M(x) / (P)	0.131	1C1	0.063	(A,B)
M(x) / (P)	0.131	2C !	0.038	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.135	4A	9.778	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.144	2A	0.030	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.121	4B	4.951	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.124	2B	0.030	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC	11.8	11.8	-11.8	-11.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-16.2	-16.2	16.2	16.2
Shear MT	172.5	172.5	172.5	172.5	172.5	172.5	172.5	172.5
Tot. Shear	184.4	184.4	160.7	160.7	156.3	156.3	188.7	188.7
Str. Int.	368.7	368.7	321.4	321.4	312.7	312.7	377.4	377.4

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		8.7	8.7	-8.7	-8.7	-10.8	-10.8	10.8	10.8
Shear Q (SUS)		169.2	169.2	169.2	169.2	169.2	169.2	169.2	169.2
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		17.3	17.3	17.3	17.3	21.7	21.7	21.7	21.7
Pm+Pl+Q (Total)		355.8	355.8	321.1	321.1	316.8	316.8	360.1	360.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	21.67	207.00	Passed
Pm+Pl+Q (TOTAL)	360.11	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N/mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		11.8	11.8	-11.8	-11.8	-16.2	-16.2	16.2	16.2
Shear Q (SUS)		172.5	172.5	172.5	172.5	172.5	172.5	172.5	172.5
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		23.7	23.7	23.7	23.7	32.3	32.3	32.3	32.3
Pm+Pl+Q (Total)		368.7	368.7	321.4	321.4	312.7	312.7	377.4	377.4

Vessel Stress Summation Comparison (N/mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	32.35	207.00	Passed
Pm+Pl+Q (TOTAL)	377.39	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	4914.0000	mm.
Skirt Thickness	STHK	14.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material	SA-36		

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm2
Basing Yield Stress		262.00	N./mm2
Inside Diameter of Basing	DI, D4	4730.0000	mm.
Outside Diameter of Basing	DOU, D5	5250.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	11.5300	cm2
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm2
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	5082.0000	mm.
Bolt Allowable Shear Stress		82.800	N./mm2
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm2
Allowable Comp. Strength of Concrete	FC	22.5	N./mm2
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	158.0000	mm.
Width of Gussets at Base Plate	BWDT	158.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm2
Height of Chair	h1	350.0000	mm.
Distance between Gussets	RG, b6	96.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	44.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	47145.9	Kgf
Operating Weight of Vessel	ROW	74213.9	Kgf
Earthquake Moment on Basing	EQMOM	62720.9	Kg-m.
Wind Moment on Basing	WIMOM	14549.6	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 62721/5082.0) - 71887) / 24$$

= -938.4 Kgf

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 14550/5082.0) - 47146) / 24$$

$$= -1487.3 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5082.0) - 74214) / 24$$

$$= -3092.2 \text{ Kgf}$$

Governing Bolt Load Condition, **** No Tensile Bolt Loads ****

Area Available in one Bolt	Abss :	11.5300	cm ²
Area Available in all the Bolts	Abss * RN :	276.7200	cm ²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
 It is possible that this method is encountering a problem. It is strongly
 recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 0.342[(2 \cdot 1.0 \cdot 5082.0 + 260.0) / (2 \cdot 1.0 \cdot 5082.0)]$$

$$= 0.35 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 168.0, b = 332.616, l/b = 0.505087

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(118.979, 318.756) = 318.756 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 318.756 / 172.9)} + 1.0$$

$$= 11.415 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 168.0, b = 96.0, b/l = 0.571429

As b/l (0.571) is less than 1, inverting b/l = 1.75.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2 \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1) - [(0.7 - g2)P / (4\pi)]]$$

$$= 16224.81 / (4 \cdot \pi) [1.3(\ln((2 \cdot 168.0 \sin(\pi \cdot 84.0 / 168.0)) / (\pi \cdot 39.688))) + 1] - [(0.7 - 0.044) \cdot 16224.81 / (4 \cdot \pi)]$$

$$= 2108.7163 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(2108.72) / 174.66)} + 1.0$$

$$= 27.6531 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot lavg) + Ca$$

$$= 16224.81 / (124.11 \cdot 158.0) + 1.0$$

$$= 10.525 \text{ (not less than } 9.525 + 1.0 \text{) mm.}$$

Bolt spacing [m]:



$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 5082.0/24 = 665.232 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76(16225 \cdot 84.0 / (665.23 \cdot 306.0 \cdot 171))^{2/3} \cdot 2457.01^{1/3} + Ca$$

$$= 12.546 + 1.0 = 13.546 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 12334.9 / (11.53 \cdot 24)$$

$$= 4.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	11.4148	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	27.6531	mm.
Actual Top Ring Thickness as entered by user	44.0000	mm.
Required Gusset thickness, + CA	10.5250	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	13.5464	mm.
Given Thickness of Skirt	14.0000	mm.
Required Gusset Height to meet local stress	258.1237	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 8.319 \cdot (14.0 - 1.0)$$

$$= 11.03 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 11 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 0.831 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 16224.8 / (162.0 + 2 \cdot (274.0 + 44.0))$$

$$= 20.332 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 16224.8 \cdot 84.0 / (162.0 \cdot (318.0) + 0.6667 \cdot (318.0)^2)$$

$$= 11.459 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$

$$= \sqrt{ 20.33^2 + 11.46^2 }$$

$$= 23.339 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 23.34 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 1.759 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 16224.8 / (2 \cdot 158.0)$$

$$= 51.344 \text{ Kg/mm.}$$

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 16224.8 • 84.0 / (2 • 318.0 • 158.0)
 = 13.563 Kgf/mm.

Resultant Weld Load [Wr]:
 = $\sqrt{(Wv^2 + Wh^2)}$
 = $\sqrt{(51.34^2 + 13.56^2)}$
 = 53.105 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 53.11 / [(0.4 • 230) • 2 • 0.707]
 = 4.002 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500 mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500 mm.
Required Top Plate to Skirt Weld Size	6.3500 mm.
Required Gusset to Top Plate Double Fillet Weld Size	4.0020 mm.

Tailing Lug Stress Calculation Results :

Tail Lug Type:	Dual		
Tail lug yield stress	Syt	262.00	N./mm
Tail lug allowable stress	Sat	138.00	N./mm
Tail lug thickness	TTL	80.0000	mm.
Pin hole diameter	DPH	130.0000	mm.
Fillet weld size	WS	30.0000	mm.
Tail lug load	TLOAD	13261	Kgf

Decay Length Along the Skirt (e):
 = $0.5 \cdot \sqrt{(DS/2 \cdot Ts)}$
 = $0.5 \cdot \sqrt{(4914.0/2 \cdot 14.0)}$
 = 92.73 mm.

Skirt Effective Length :
 = HG + TTA + e
 = 274.0 + 44.0 + 92.73
 = 410.73 mm.

Section	CG (Y) mm.	Width (H) mm.	Area (A) cm ²	Moment Area mm. ³
Top plate, t	176.00	168.00	73.92	1300992.00
Skirt, s	85.00	410.73	57.50	488772.84
Bottom plate, b	130.00	260.00	83.20	1081600.00
TOTAL			214.62	2871364.75

Calculated Overall CG (Y):
 = Mtotal / Atotal
 = 2871365/214.62
 = 133.79 mm.

Calculated Moment of Inertia (I):
 = (At • Yt² + Topwth³ • TTA/12 + As • Ys² +
 Ts³ • Hs/12 + Ab • Yb² + Hb³ • Tb/12) -
 Mtotal²/Atotal
 = (73.92 • 176.0² + 168.0³ • 44.0/12 + 57.5 • 85.0² +
 14.0³ • 410.73/12 + 83.2 • 130.0² + 260.0³ • 32.0/12) -
 2871365²/214.62
 = 9132.73 cm⁴

$$\alpha = \arcsin(OS / (DS / 2))$$

$$= \arcsin(500.0/(4914.0/2))$$

$$= 0.205 \text{ rad. (11.742 deg.)}$$

$$K = \sin(\alpha) \cdot (\sin(\alpha) - \pi + \alpha) + k2 \cdot (1 + \cos(\alpha))$$

$$= \sin(0.2) \cdot (\sin(0.2) - 3.14 + 0.2) + 1.0 \cdot (1 + \cos(0.2))$$

$$= 1.42$$

Required Section Modulus (Zr) - per Roark's Table 17 ref 4

$$= K \cdot TLOAD \cdot (DI/2 + Y) / (0.66 \cdot Sy \cdot \pi)$$

$$= 1.42 \cdot 13261.84 \cdot (4730.0/2 + 133.79) / (0.66 \cdot 262.0 \cdot 3.142)$$

$$= 851219.38 \text{ mm.}^3$$

Calculated Section Modulus (Zc):

$$= I / Y$$

$$= 9132.73/133.79$$

$$= 682634.31 \text{ mm.}^3 \text{ ...[Insufficient section modulus!]}$$

Allowable Shear at the Pin Hole (Ssa):

$$= 0.4 \cdot Syt$$

$$= 0.4 \cdot 262.0$$

$$= 104.80 \text{ N./mm}^2$$

Tail Lug Radius (R):

$$= (TBA + HG + TTA) / 2$$

$$= (32.0 + 274.0 + 44.0) / 2$$

$$= 175.00 \text{ mm.}$$

Calculated Shear at the Pin Hole (Ssc):

$$= TLOAD / (2 \cdot TTL \cdot (R - DPH / 2))$$

$$= 13261.8 / (2 \cdot 80.0 \cdot (175.0 - 130.0/2))$$

$$= 7.39 \text{ N./mm}$$

Allowable Bearing Stress at the Pin Hole (Sba):

$$= 0.75 \cdot Syt$$

$$= 0.75 \cdot 262.0$$

$$= 196.50 \text{ N./mm}^2$$

Calculated Bearing Stress at the Pin Hole (Sbc):

$$= TLOAD / (DPH \cdot TTL)$$

$$= 13261.8 / (130.0 \cdot 80.0)$$

$$= 12.51 \text{ N./mm}^2$$

Weld Length (LW):

$$= 2 \cdot (TOPWTH + TBA + TTA + HG + (DOU - DS) / 2)$$

$$= 2 \cdot (168.0 + 32.0 + 44.0 + 274.0 + (5250.0 - 4914.0)/2)$$

$$= 1371.60 \text{ mm.}$$

Allowable Weld Stress (Swa):

$$= 0.49 \cdot Sat$$

$$= 0.49 \cdot 138.0$$

$$= 67.62 \text{ N./mm}^2$$

Calculated Weld Stress (Swc):

$$= TLOAD / (0.707 \cdot WS \cdot LW)$$

$$= 13261.8 / (0.707 \cdot 30.0 \cdot 1371.6)$$

$$= 4.47 \text{ N./mm}^2$$

Tailing Lug Results Summary (mm.³ & N./mm²):

	Actual	Allowable or Required	Pass/Fail
Section Modulus	682634.31	851219.38	Failed
Shear Stress	7.39	104.80	Passed
Bearing Stress	12.51	196.50	Passed
Weld Stress	4.47	67.62	Passed

Review notes about nozzle loadings and supports in the Warnings report.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 81 of 152

Basing Calculations: Step: 26 2:23pm Sep 16,2026



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Input, Nozzle Desc: D1-2"

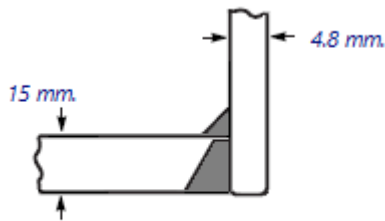
From: 30

Pressure for Reinforcement Calculations	P	6.182	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	4881.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	15.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316L	
Material UNS Number		S31603	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	115.00	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80S	
Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	15.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2800.0	mm.
Centerline Distance to Tangent		1600.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: D1-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.993 in.
Actual Thickness Used in Calculation	0.191 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.18 \cdot 4883.0 \cdot 0.8996 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 6.18)$
 $= 9.8444 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.18 \cdot 25.32 / (115 \cdot 1.0 \cdot 0.6 \cdot 6.18)$
 $= 0.1365 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.2698 mm.
Parallel to Vessel Wall, opening length	d	50.6349 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.1126 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: D1-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1365 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.9365 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.9365 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 3.4200 mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.42, \max(11.9365, 2.5)]$
 $= 3.4200 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(0.1365, 3.42)$
 $= 3.4200 \text{ mm.}$

Available Nozzle Neck Thickness = 4.8450 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	51.5,	Allowable	:	115.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	236.0 N./mm2	Passed
Occasional	:	1.5,	Allowable	:	152.9 N./mm2	Passed

Shear : 34.8, Allowable : 80.5 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: D1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 4.8450 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.3915 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1031 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 165.1030 mm.

Input Echo, WRC107/537 Item 1, Description: D1-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	8785.800	mm.
Vessel Thickness	Tv	15.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	50.635	mm.
Nozzle Thickness	Tn	4.845	mm.
Nozzle Material		SA-312 TP316L	
Nozzle UNS Number		S31603	
Nozzle Cold S.I. Allowable	SNmc	115.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	115.00	N./mm ²
Design Internal Pressure	Dp	6.182	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

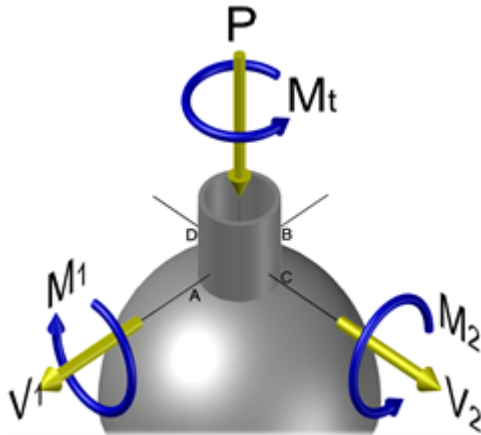
Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
--	----

Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1
 Warning - The ratio Ro/Rm, 0.0069, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(4400.9 (15.0 - 1.0))}$$

$$= 879.447 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

Dimensionless Parameters: U = 0.12 TAU = 5.73 RHO = 2.89

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.14268	(A,B,C,D)
$M(x) / P$	SP 3	0.10098	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.51621	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.40121	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.51621	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.40121	(A,B,C,D)
$N(y) * T / P$	SP 3	0.26189	(A,B,C,D)
$M(y) / P$	SP 3	0.21038	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.09156	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	1.21494	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.09156	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	1.21494	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-2.7	-2.7	-2.7	-2.7	-2.7	-2.7	-2.7	-2.7
Rad.	Bend. P	-11.6	11.6	-11.6	11.6	-11.6	11.6	-11.6	11.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-3.7	-3.7	3.7	3.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-17.3	17.3	17.3	-17.3
Rad.	Memb. ML	-3.7	-3.7	3.7	3.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-17.3	17.3	17.3	-17.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-35.3	22.5	6.7	-4.7	-35.3	22.5	6.7	-4.7
Tang.	Memb. P	-5.0	-5.0	-5.0	-5.0	-5.0	-5.0	-5.0	-5.0
Tang.	Bend. P	-24.2	24.2	-24.2	24.2	-24.2	24.2	-24.2	24.2
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-52.4	52.4	52.4	-52.4
Tang.	Memb. ML	-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-52.4	52.4	52.4	-52.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-82.2	70.9	23.9	-32.6	-82.2	70.9	23.9	-32.6
Shear VC		1.9	1.9	-1.9	-1.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Shear MT		6.2	6.2	6.2	6.2	6.2	6.2	6.2	6.2
Tot. Shear		8.1	8.1	4.4	4.4	4.4	4.4	8.1	8.1
Str. Int.		83.6	72.2	25.0	33.3	82.7	71.3	27.2	34.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	97.0	97.0	97.0	97.0	97.0	97.0	97.0	97.0
Rad.	Pl (SUS)	-6.4	-6.4	1.0	1.0	-6.4	-6.4	1.0	1.0
Rad.	Q (SUS)	-28.9	28.9	5.7	-5.7	-28.9	28.9	5.7	-5.7
Long.	Pm (SUS)	97.0	97.0	97.0	97.0	97.0	97.0	97.0	97.0
Long.	Pl (SUS)	-5.7	-5.7	-4.4	-4.4	-5.7	-5.7	-4.4	-4.4
Long.	Q (SUS)	-76.6	76.6	28.3	-28.3	-76.6	76.6	28.3	-28.3
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.9	1.9	-1.9	-1.9	-1.9	-1.9	1.9	1.9
Shear	Q (SUS)	6.2	6.2	6.2	6.2	6.2	6.2	6.2	6.2
Pm (SUS)		97.0	97.0	97.0	97.0	97.0	97.0	97.0	97.0
Pm+Pl (SUS)		92.9	92.9	98.6	98.6	92.9	92.9	98.6	98.6
Pm+Pl+Q (Total)		63.0	169.3	122.0	92.9	62.1	168.3	124.2	94.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	97.01	138.00	Passed
Pm+Pl (SUS)	98.60	207.00	Passed
Pm+Pl+Q (TOTAL)	169.25	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 87 of 152

Nozzle Calcs.: D1-2" Nozl: 29 2:23pm Sep 16,2026



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Input, Nozzle Desc: K1-2"

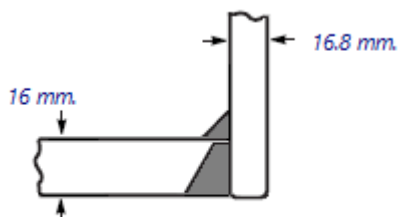
From: 40

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.6436 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.6436 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.6436, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 90 of 152

Nozzle Calcs.: K1-2" Nozl: 30 2:23pm Sep 16,2026



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: K3-2"

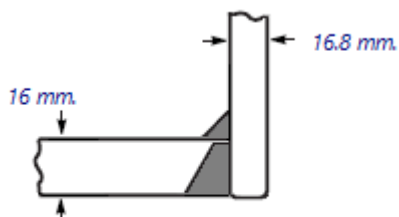
From: 40

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		80.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.6436 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.6436 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.6436, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 93 of 152

Nozzle Calcs.: K3-2" Nozl: 31 2:23pm Sep 16,2026



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: K5-2"

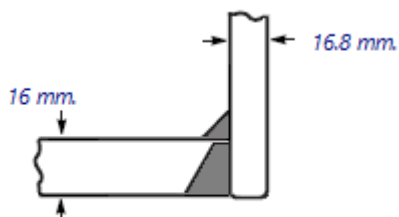
From: 40

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.6436 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.6436 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.6436, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 96 of 152

Nozzle Calcs.: K5-2" Nozl: 32 2:23pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: K7-2"

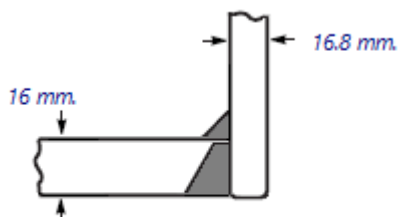
From: 40

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K7-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.6436 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.6436 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.6436, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 99 of 152

Nozzle Calcs.: K7-2" Nozl: 33 2:23pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: K2-2"

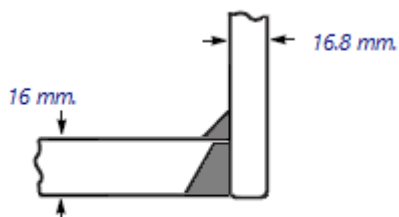
From: 40

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		80.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 10.7301 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.7301 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.7301 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.7301, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 102 of 152

Nozzle Calcs.: K2-2" Nozl: 34 2:23pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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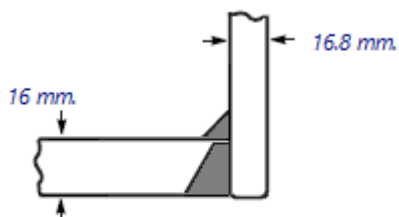
Input, Nozzle Desc: K4-2"
From: 40

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 10.7301 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.7301 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.7301 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.7301, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 105 of 152

Nozzle Calcs.: K4-2" Nozl: 35 2:23pm Sep 16,2026



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: K6-2"

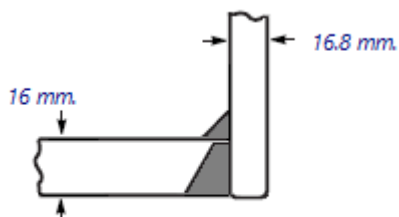
From: 40

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 10.7301$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.7301	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.7301	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.7301, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.0000	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 108 of 152

Nozzle Calcs.: K6-2" Nozl: 36 2:23pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: K8-2"

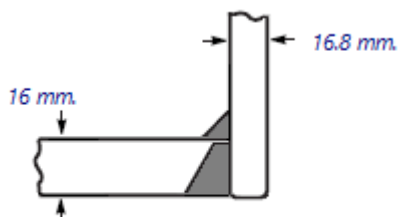
From: 40

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K8-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 10.7301 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.7301 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.7301 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.7301, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K8-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 111 of 152

Nozzle Calcs.: K8-2" Nozl: 37 2:23pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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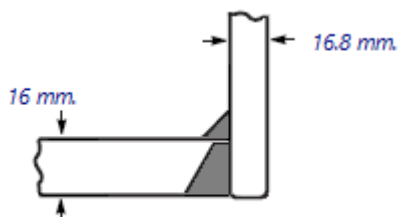

Input, Nozzle Desc: K10-2"
From: 40

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K10-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 10.7301$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K10-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.7301	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.7301	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(11.7301, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K10-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

Process Air 1st KOD 196-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 1st KO Drum-196-D -7% Page 114 of 152

Nozzle Calcs.: K10-2" Nozl: 38 2:23pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.7393 mm.

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Input, Nozzle Desc: N1-36"

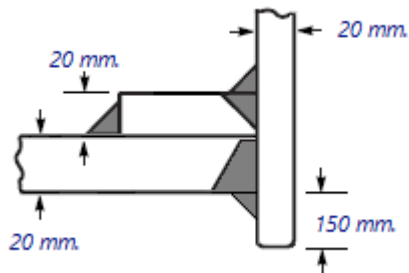
From: 60

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3100.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	150.0000	mm.
Weld leg size, Inside Element to Shell	Wi	14.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1614.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		350.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: N1-36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.0 \cdot 457.2 / (138 \cdot 1.0 + 0.4 \cdot 6.0)$
 $= 1.9845 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1760.8000 mm.
Parallel to Vessel Wall, opening length	d	880.4000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm.
Normal to Vessel Wall, Inward		35.0000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

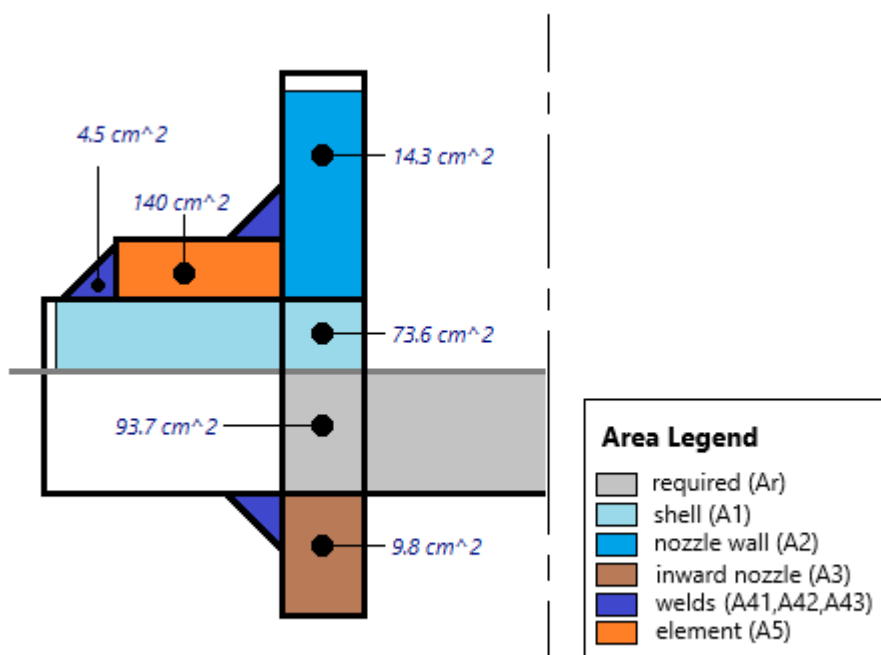


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	93.706	NA	NA
Area in Shell	A1	73.570	NA	NA
Area in Nozzle Wall	A2	14.265	NA	NA
Area in Inward Nozzle	A3	9.800	NA	NA
Area in Welds	A41+A42+A43	4.512	NA	NA
Area in Element	A5	140.000	NA	NA
TOTAL AREA AVAILABLE	Atot	242.147	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (880.4 \cdot 10.6436 \cdot 1.0 + 2 \cdot 17.0 \cdot 10.6436 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 93.706 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 880.4 (1.0 \cdot 19.0 - 1.0 \cdot 10.644) - 2 \cdot 17.0$$

$$(1.0 \cdot 19.0 - 1.0 \cdot 10.6436) \cdot (1 - 1.0)$$

$$= 73.570 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 47.5) \cdot (17.0 - 1.98) \cdot 1.0$$

$$= 14.265 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$



$$= 2 \cdot 14.0 \cdot (35.0) \cdot 1.0$$

$$= 9.800 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (9.7573)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$$

$$= 4.512 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (1614.4 - 914.4) \cdot 20.0 \cdot 1.0$$

$$= 140.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.9845 \text{ mm.}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm.}$

Wall Thickness, shell/head, internal pressure $trb_1 = 11.6436 \text{ mm.}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 11.6436 \text{ mm.}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm.}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(11.6436, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.9845, 11.3312)$$

$$= 11.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	24.2,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	320.8 N./mm ²	Passed
Occasional	:	7.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	10.5,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 1.984$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.984$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.984$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
--	-------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20.0, tr = 10.644, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 10.644, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -33 °C
 Gov. MDMT of the Reinforcement Pad : -33 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -33 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.00/19.60 = 0.306$

Weld Size Calculations, Description: N1-36"

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.
 Intermediate Calc. for Inward Weld TminIns 17.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (93.7063 - 73.5697 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 10.6436)) \cdot 138)$
 $= 32333.86 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (14.2647 + 140.0 + 4.5121 - 95.2053 \cdot 1.0) \cdot 138$
 $= 222088.19 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (14.2647 + 9.8 + 1.9521 + (6.46)) \cdot 138$
 $= 45700.76 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (14.2647 + 9.8 + 4.5121 + 140.0 + (6.46)) \cdot 138$
 $= 246308.67 \text{ Kg}$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 99038. \text{ Kg}$$

Shear, Inward Nozzle Weld [Sinw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 9.7573 \cdot 0.49 \cdot 138$$

$$= 96635. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1614.4 \cdot 16.0 \cdot 0.49 \cdot 138$$

$$= 279768. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 448.7) \cdot (20.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 236050. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 914.4 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 299136. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 239309. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (279768 + 236050) = 515817 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (99038 + 299136 + 239309 + 96635) = 734117 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (279768 + 239309 + 96635) = 615711 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 515817 Kg, must exceed W = 32333 Kg	or W1 = 222088 Kg
Path 2-2 = 734117 Kg, must exceed W = 32333 Kg	or W2 = 45700 Kg
Path 3-3 = 615710 Kg, must exceed W = 32333 Kg	or W3 = 246308 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 43.2080 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 469.9999 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.236 %

Input Echo, WRC297 Item 1, Description: N1-36" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.0000 mm.
Shell External Corrosion Allowance	0.0000 mm.



Shell Diameter	4881.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	914.400	mm.
Nozzle Thickness	20.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1614.400	mm.
Thickness of Reinforcing Pad	20.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	6.00	bars
Radial Load	P	6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.91	Kg-m.
Longitudinal Moment	Ml	11559.91	Kg-m.
Torsional Moment	Mt	16514.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(2461.0(20.0 - 1.0))}$
 $= 1627.986 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4922.000	mm.
Nozzle Outside Diameter	(d) =	914.400	mm.
Vessel Thickness used	(T) =	39.000	mm.
Nozzle Thickness used	(t) =	17.000	mm.
T / t	=	2.294	
d / t	=	53.788	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.087$$

Nr/P	=	0.043
Mr/P	=	0.080
M0/P	=	0.048
N0/P	=	0.159
MrD/Mc	=	0.175
NrDL/Mc	=	0.086
M0d/Mc	=	0.124
N0DL/Mc	=	0.233
MrD/Ml	=	0.093



NrDL/M1 = 0.051

MØD/M1 = 0.056

NØDL/M1 = 0.317

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	37	37	37	37
Outplane Stress Summary	-35	45	70	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-45	45	45	-45
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	-53	79	44	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	54	79	70	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18
Outplane Stress Summary	-80	65	78	-17

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
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(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-85	85	85	-85
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	37	37	37	37
Inplane Stress Summary	-78	135	106	-21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	82	135	107	23

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19
Hoop Stress Summary	-14	-13	37	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-81	81	-81	81
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-172	172	172	-172
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-257	251	109	-72

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7



Shear Stress Summary	9	9	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	258	266	109	110

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19

Hoop Stress Summary	-7	-7	30	30
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-81	81	-81	81
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-339	339	339	-339
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-424	418	275	-239
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	424	425	276	270

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(2451.0(20.0 - 1.0))}$$

$$= 1626.535 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 4902.000 mm.

Nozzle Outside Diameter (d) = 1614.400 mm.

Vessel Thickness used (T) = 19.000 mm.

Nozzle Thickness used (t) = 367.000 mm.

T / t = 0.052



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.290$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.023
MrD/Mc	=	0.238
NrDL/Mc	=	0.101
M0d/Mc	=	0.070
N0DL/Mc	=	0.045
MrD/M1	=	0.045
NrDL/M1	=	0.047
M0D/M1	=	0.014
N0DL/M1	=	0.021

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-21	21	-21	21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	76	77	76	77
Outplane Stress Summary	31	106	71	82

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-79	79	-79	79
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	-111	153	12	65

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	143	153	71	82

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-21	21	-21	21
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-82	82	82	-82
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38

Outplane Stress Summary	-77	128	103	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-79	79	-79	79
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-277	277	277	-277
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	77	76	77

Inplane Stress Summary	-308	406	285	-109
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	308	406	285	109
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	37	37	37	37	37	37	37	37
Circ. Pl (SUS)	-32	-32	18	18	-7	-7	5	5
Circ. Q (SUS)	-39	39	15	-15	-106	106	64	-64
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-5	-5	3	3	-25	-25	11	11
Long. Q (SUS)	-66	66	24	-24	-72	72	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0
Pm+Pl (SUS)	13.0	13.0	55.0	55.0	37.0	37.0	42.0	42.0

Pm+Pl+Q (Total)	53.5	79.3	70.4	43.4	80.9	136.1	106.3	23.9
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	37.00	138.00	Passed
Pm+Pl (SUS)	55.00	207.00	Passed
Pm+Pl+Q (TOTAL)	136.13	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	77	76	77	76	77	76	77
Circ. Pl (SUS)	-8	-8	0	0	-27	-27	11	11
Circ. Q (SUS)	-37	37	-5	5	-356	356	198	-198
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	-17	-17	1	1	-12	-12	4	4
Long. Q (SUS)	-131	131	-27	27	-103	103	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	76.0	77.0	76.0	77.0	76.0	77.0	76.0	77.0
Pm+Pl (SUS)	68.0	69.0	76.0	77.0	49.0	50.0	87.0	88.0
Pm+Pl+Q (Total)	141.1	152.1	71.1	82.2	307.0	406.0	285.0	110.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	138.00	Passed
Pm+Pl (SUS)	88.00	207.00	Passed
Pm+Pl+Q (TOTAL)	406.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-32	-32	18	18	-25	-25	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-11	-11	9	9	-11	-11	9	9
Long. Q (SUS)	-253	253	91	-91	-420	420	258	-258



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	14.1	13.1	36.1	37.1	7.2	6.2	29.1	30.1
Pm+Pl+Q (Total)	255.3	264.5	109.5	110.7	422.1	424.2	276.3	270.5

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	37.05	207.00	Passed
Pm+Pl+Q (TOTAL)	424.17	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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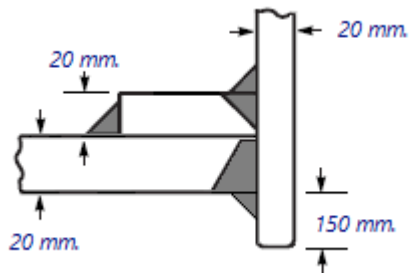
**Input, Nozzle Desc: N2-36"****From: 60**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3500.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	150.0000	mm.
Weld leg size, Inside Element to Shell	Wi	14.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1614.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		350.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: N2-36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.0 \cdot 457.2 / (138 \cdot 1.0 + 0.4 \cdot 6.0)$
 $= 1.9845 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1760.8000	mm.
Parallel to Vessel Wall, opening length	d	880.4000	mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000	mm.
Normal to Vessel Wall, Inward		35.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

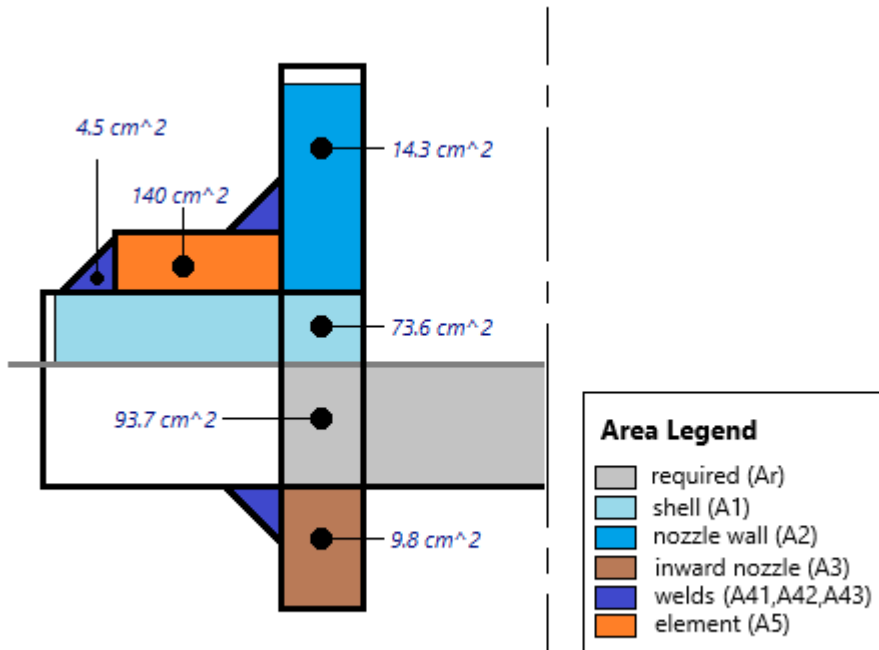


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	93.706	NA	NA
Area in Shell	A1	73.570	NA	NA
Area in Nozzle Wall	A2	14.265	NA	NA
Area in Inward Nozzle	A3	9.800	NA	NA
Area in Welds	A41+A42+A43	4.512	NA	NA
Area in Element	A5	140.000	NA	NA
TOTAL AREA AVAILABLE	Atot	242.147	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (880.4 \cdot 10.6436 \cdot 1.0 + 2 \cdot 17.0 \cdot 10.6436 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 93.706 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 880.4 (1.0 \cdot 19.0 - 1.0 \cdot 10.644) - 2 \cdot 17.0$$

$$(1.0 \cdot 19.0 - 1.0 \cdot 10.6436) \cdot (1 - 1.0)$$

$$= 73.570 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 47.5) \cdot (17.0 - 1.98) \cdot 1.0$$

$$= 14.265 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$



$$= 2 \cdot 14.0 \cdot (35.0) \cdot 1.0$$

$$= 9.800 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - c_n / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$

$$= 10.0^2 \cdot 1.0 + (9.7573)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$$

$$= 4.512 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr4$$

$$= (1614.4 - 914.4) \cdot 20.0 \cdot 1.0$$

$$= 140.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.9845 \text{ mm.}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm.}$

Wall Thickness, shell/head, internal pressure $trb1 = 11.6436 \text{ mm.}$

Wall Thickness $tb1 = \max(trb1, tr_{16b}) = 11.6436 \text{ mm.}$

Wall Thickness $tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm.}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(11.6436, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.9845, 11.3312)$$

$$= 11.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 24.2,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 320.8 N./mm ²	Passed
Occasional	: 7.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 10.5,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 1.984$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.984$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.984$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20.0, tr = 10.644, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 10.644, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -33 °C
 Gov. MDMT of the Reinforcement Pad : -33 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -33 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.00/19.60 = 0.306$

Weld Size Calculations, Description: N2-36"

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.
 Intermediate Calc. for Inward Weld TminIns 17.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (93.7063 - 73.5697 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 10.6436)) \cdot 138)$
 $= 32333.86 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (14.2647 + 140.0 + 4.5121 - 95.2053 \cdot 1.0) \cdot 138$
 $= 222088.19 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (14.2647 + 9.8 + 1.9521 + (6.46)) \cdot 138$
 $= 45700.76 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (14.2647 + 9.8 + 4.5121 + 140.0 + (6.46)) \cdot 138$
 $= 246308.67 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 99038. \text{ Kg}$$

Shear, Inward Nozzle Weld [Sinw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 9.7573 \cdot 0.49 \cdot 138$$

$$= 96635. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1614.4 \cdot 16.0 \cdot 0.49 \cdot 138$$

$$= 279768. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 448.7) \cdot (20.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 236050. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 914.4 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 299136. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 239309. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (279768 + 236050) = 515817 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (99038 + 299136 + 239309 + 96635) = 734117 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (279768 + 239309 + 96635) = 615711 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 515817 Kg, must exceed W = 32333 Kg	or W1 = 222088 Kg
Path 2-2 = 734117 Kg, must exceed W = 32333 Kg	or W2 = 45700 Kg
Path 3-3 = 615710 Kg, must exceed W = 32333 Kg	or W3 = 246308 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 43.2080 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 469.9999 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 2.236 \%$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 494.15 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 880.40 mm.

Input Echo, WRC297 Item 1, Description: N2-36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	4881.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	914.400	mm.
Nozzle Thickness	20.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1614.400	mm.
Thickness of Reinforcing Pad	20.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	6.00	bars
Radial Load	P	6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.91	Kg-m.
Longitudinal Moment	Ml	11559.91	Kg-m.
Torsional Moment	Mt	16514.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(2461.0(20.0 - 1.0))}$
 $= 1627.986 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4922.000	mm.
Nozzle Outside Diameter	(d) =	914.400	mm.
Vessel Thickness used	(T) =	39.000	mm.
Nozzle Thickness used	(t) =	17.000	mm.
T / t	=	2.294	
d / t	=	53.788	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.087$$

Nr/P	=	0.043
Mr/P	=	0.080
M0/P	=	0.048
N0/P	=	0.159
MrD/Mc	=	0.175



NrDL/Mc = 0.086
 MØd/Mc = 0.124
 NØDL/Mc = 0.233
 MrD/Ml = 0.093
 NrDL/Ml = 0.051
 MØD/Ml = 0.056
 NØDL/Ml = 0.317

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	37	37	37	37
Outplane Stress Summary	-35	45	70	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-45	45	45	-45
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	-53	79	44	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	54	79	70	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18



Outplane Stress Summary -80 65 78 -17

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du D1
 (Stresses parallel to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-85	85	85	-85
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	37	37	37	37

Inplane Stress Summary -78 135 106 -21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du D1
 (Shear stress normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary 2 2 3 3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du D1
 (Stress Intensities Left Left Right Right
 Outside Inside Outside Inside

Two * Max Shear Stress 82 135 107 23

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE Au Al Bu B1
 (Stresses in the Top Top Bottom Bottom
 hoop direction) Outside Inside Outside Inside

Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19

Hoop Stress Summary -14 -13 37 37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE Au Al Bu B1
 (Stresses Normal to Top Top Bottom Bottom
 pipe cross-section) Outside Inside Outside Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-81	81	-81	81
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-172	172	172	-172
Axial Pressure Stress	9	9	9	9

Axial Stress Summary -257 251 109 -72

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE Au Al Bu B1
 (Shear stress) Outside Inside Outside Inside



Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	258	266	109	110
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19

Hoop Stress Summary	-7	-7	30	30
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-81	81	-81	81
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-339	339	339	-339
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-424	418	275	-239
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	424	425	276	270
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(2451.0(20.0 - 1.0))}$$

$$= 1626.535 \text{ mm.}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 4902.000 \text{ mm.}$$



Nozzle Outside Diameter (d) = 1614.400 mm.
 Vessel Thickness used (T) = 19.000 mm.
 Nozzle Thickness used (t) = 367.000 mm.
 T / t = 0.052
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.290$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.046
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.023
 MrD/Mc = 0.238
 NrDL/Mc = 0.101
 M0d/Mc = 0.070
 N0DL/Mc = 0.045
 MrD/Ml = 0.045
 NrDL/Ml = 0.047
 M0D/Ml = 0.014
 N0DL/Ml = 0.021

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-21	21	-21	21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	76	77	76	77
Outplane Stress Summary	31	106	71	82

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-79	79	-79	79
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	-111	153	12	65

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 143 153 71 82

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-21	21	-21	21
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-82	82	82	-82
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	-77	128	103	-18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-79	79	-79	79
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-277	277	277	-277
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	77	76	77
Inplane Stress Summary	-308	406	285	-109

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	308	406	285	109

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	37	37	37	37	37	37	37	37
Circ. Pl (SUS)	-32	-32	18	18	-7	-7	5	5
Circ. Q (SUS)	-39	39	15	-15	-106	106	64	-64
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-5	-5	3	3	-25	-25	11	11
Long. Q (SUS)	-66	66	24	-24	-72	72	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	37.0	37.0	37.0	37.0	37.0	37.0	37.0	37.0
Pm+Pl (SUS)	13.0	13.0	55.0	55.0	37.0	37.0	42.0	42.0
Pm+Pl+Q (Total)	53.5	79.3	70.4	43.4	80.9	136.1	106.3	23.9

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	37.00	138.00	Passed
Pm+Pl (SUS)	55.00	207.00	Passed
Pm+Pl+Q (TOTAL)	136.13	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm2):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	77	76	77	76	77	76	77
Circ. Pl (SUS)	-8	-8	0	0	-27	-27	11	11
Circ. Q (SUS)	-37	37	-5	5	-356	356	198	-198
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	-17	-17	1	1	-12	-12	4	4
Long. Q (SUS)	-131	131	-27	27	-103	103	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	76.0	77.0	76.0	77.0	76.0	77.0	76.0	77.0
Pm+Pl (SUS)	68.0	69.0	76.0	77.0	49.0	50.0	87.0	88.0
Pm+Pl+Q (Total)	141.1	152.1	71.1	82.2	307.0	406.0	285.0	110.0

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	138.00	Passed
Pm+Pl (SUS)	88.00	207.00	Passed
Pm+Pl+Q (TOTAL)	406.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-32	-32	18	18	-25	-25	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-11	-11	9	9	-11	-11	9	9
Long. Q (SUS)	-253	253	91	-91	-420	420	258	-258
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	14.1	13.1	36.1	37.1	7.2	6.2	29.1	30.1
Pm+Pl+Q (Total)	255.3	264.5	109.5	110.7	422.1	424.2	276.3	270.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	37.05	207.00	Passed
Pm+Pl+Q (TOTAL)	424.17	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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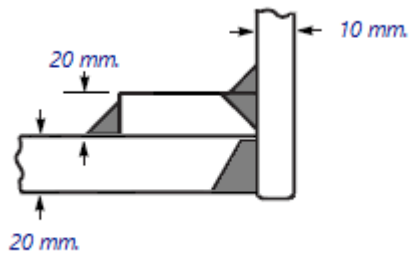
**Input, Nozzle Desc: M1-24"****From: 60**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2650.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	909.6000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		150.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.0 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 6.0)$
 $= 1.3230 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1191.2001 mm.
Parallel to Vessel Wall, opening length	d	595.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

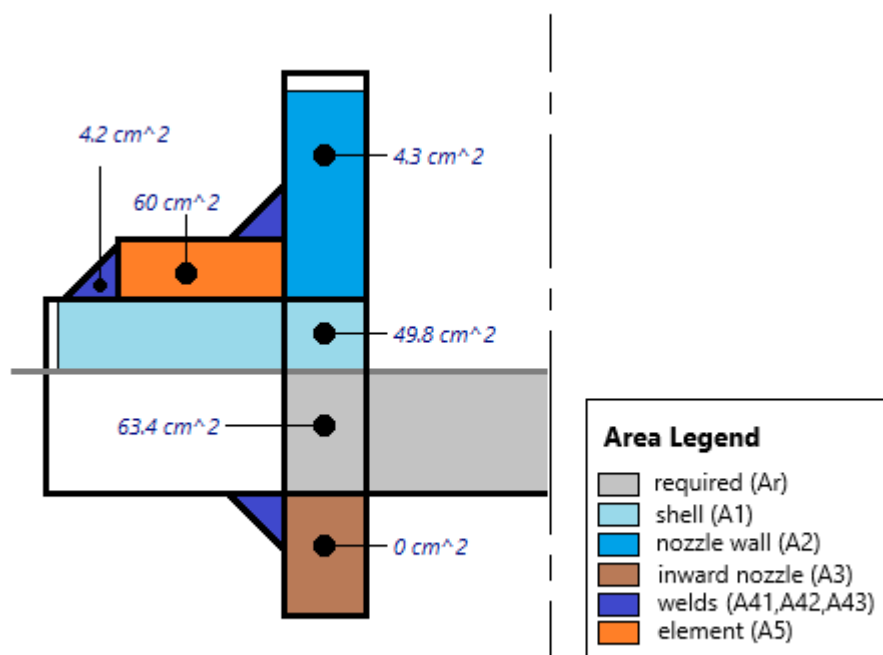


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	63.393	NA	NA
Area in Shell	A1	49.771	NA	NA
Area in Nozzle Wall	A2	4.258	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	60.000	NA	NA
TOTAL AREA AVAILABLE	Atot	118.268	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 635.2242 20.0000 mm.
 Based on given Pad Diameter: 909.6000 1.7083 mm.
 Based on Shell or Nozzle Thickness: 660.8484 10.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (595.6 \cdot 10.6436 \cdot 1.0 + 2 \cdot 7.0 \cdot 10.6436 \cdot 1.0 \cdot (1 - 1.0))$
 $= 63.393 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 595.6 (1.0 \cdot 19.0 - 1.0 \cdot 10.644) - 2 \cdot 7.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 10.6436) \cdot (1 - 1.0)$
 $= 49.771 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$

$$= (2 \cdot 37.5) \cdot (7.0 - 1.32) \cdot 1.0$$

$$= 4.258 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$$

$$= 4.240 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (909.6 - 609.6) \cdot 20.0 \cdot 1.0$$

$$= 60.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.3230 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.3230 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 10.644$, $c = 1.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -33 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -33 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6.00/19.60 = 0.306$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 7.0000 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (63.3933 - 49.7707 + 2 \cdot 7.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 10.6436)) \cdot 138)$
 $= 20815.73 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.2578 + 60.0 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 96388.83 \text{ Kg}$

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.2578 + 0.0 + 1. + (2.66)) \cdot 138$
 $= 11141.72 \text{ Kg}$

Weld Load [W3]:

$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.2578 + 0.0 + 4.24 + 60.0 + (2.66)) \cdot 138$
 $= 100131.94 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kg}$

Shear, Pad Element Weld [Spew]:

$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 909.6 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 177333. \text{ Kg}$

Shear, Nozzle Wall [Snw]:

$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 301.3) \cdot (10.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 65267. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:

$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 199424. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:

$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$

= 179481. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (177333 + 65267) = 242600 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (66025 + 199424 + 179481 + 0) = 444931 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (177333 + 179481 + 0) = 356814 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 242600 Kgf, must exceed W = 20815 Kgf	or W1 = 96388 Kgf
Path 2-2 = 444930 Kgf, must exceed W = 20815 Kgf	or W2 = 11141 Kgf
Path 3-3 = 356814 Kgf, must exceed W = 20815 Kgf	or W3 = 100131 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 19.1083 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.1083 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.668 %

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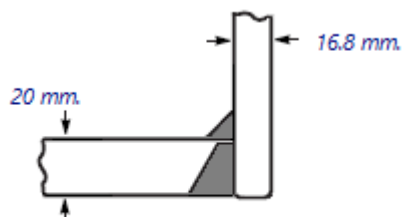

Input, Nozzle Desc: K9-2"
From: 60

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4881.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K9-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 2441.5 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 10.6436 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([6.0 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1150 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	122.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	61.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K9-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1150 mm.
Wall Thickness per UG16(b),	tr16b = 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 11.6436 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 11.6436 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm.
Wall Thickness per table UG-45	tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(11.6436, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.115, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.764, tr = 0.115, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.007$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: K9-2"



Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3643 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.7393 mm.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Tailing Lug Failure!**

Please review all reports carefully!

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**PV Elite® Pressure Vessel Design Notes**

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: